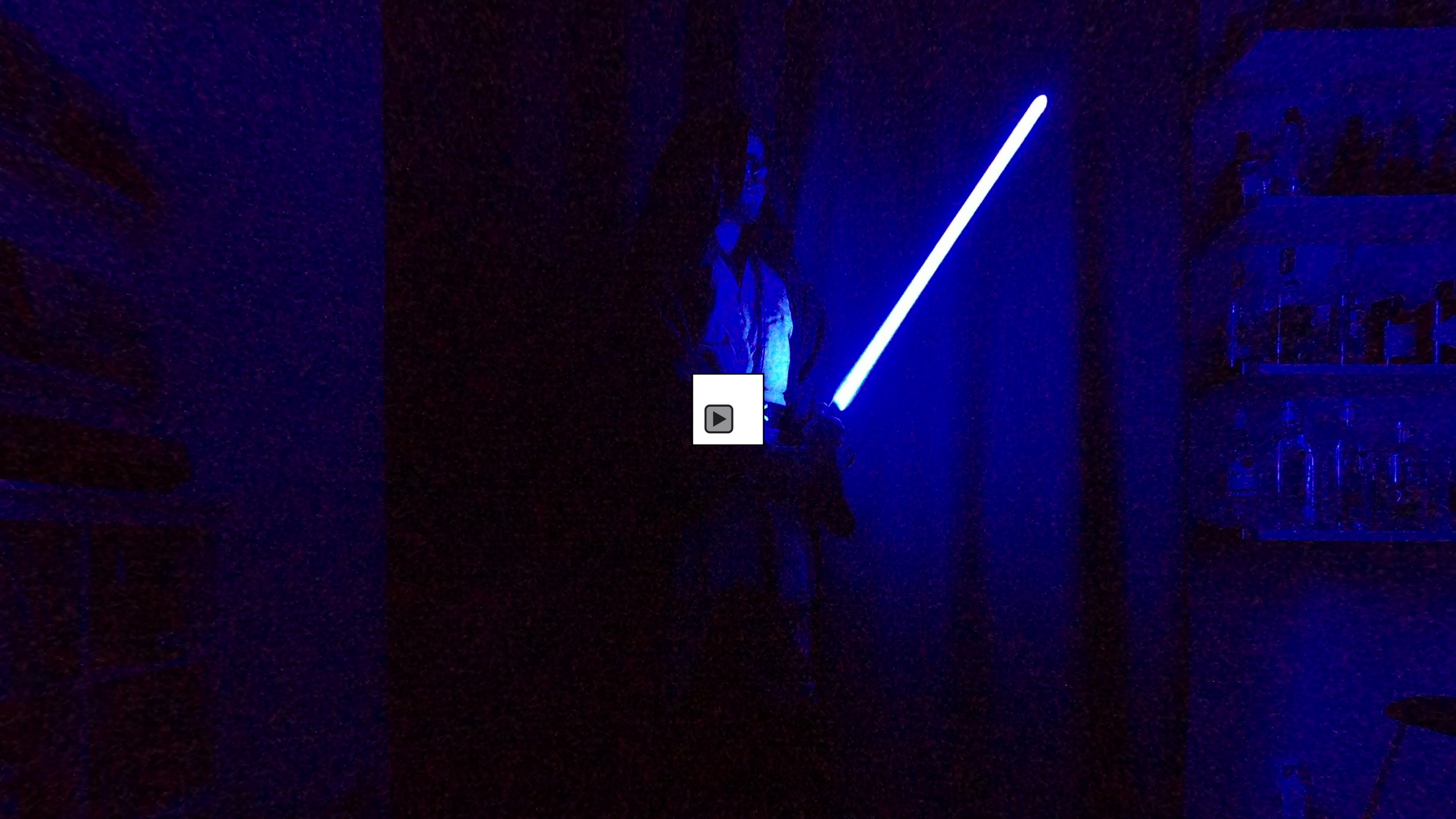
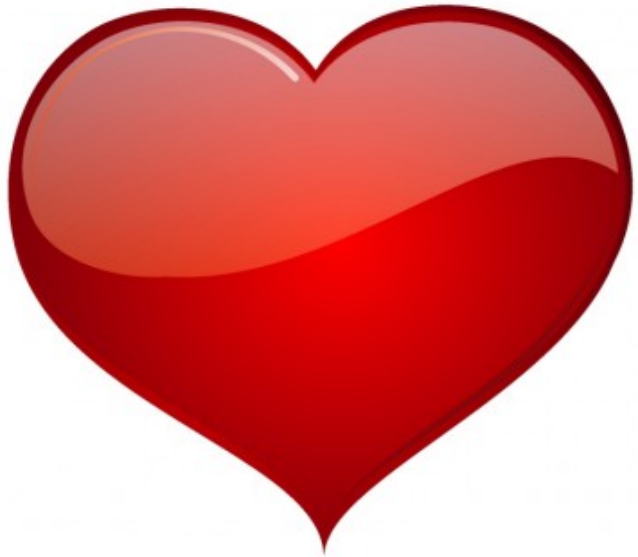
A long time ago, in a galaxy far,
far away....





http://www.

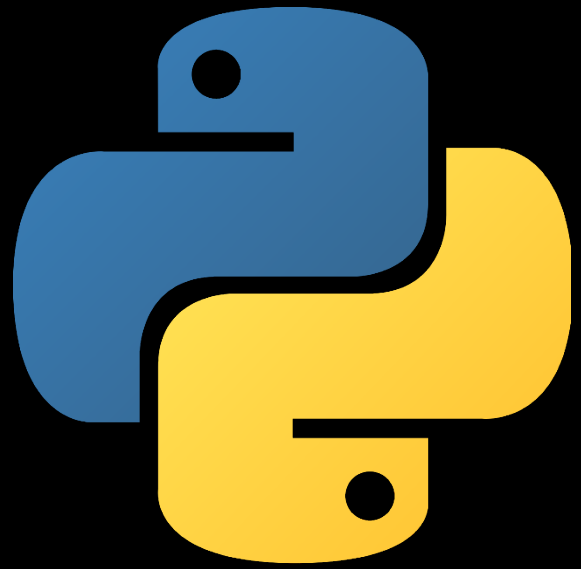
likationer

populära tillämpningarna för Python. Det passar bra som fördjupning efter programmeringen i modul 1-5.

rk och bibliotek webbprogrammering med Python. Alla delar det liga uppgifter samt att ge en struktur att bygga sitt program efter.

ket på grund av enkelheten i att installera och komma igång. Det snabbaste om deras webbplats till din projektmapp. Därefter kan du använda Bottle

kt, kan även installeras globalt på systemet med hjälp av så kallade



Episode X: The Python Way Of The Force

May the web be with you

“This is a new day, a new beginning.”

- AHSOKA TANO,
STAR WARS REBELS: "FIRE ACROSS THE GALAXY"

“This is a new day, a new beginning.”
- AHSOKA TANO,
STAR WARS REBELS: "FIRE ACROSS THE GALAXY"

libbelit.pythonanywhere.com

STAR WARS - VOTE

Please choose side

Empire

Rebels

Vote

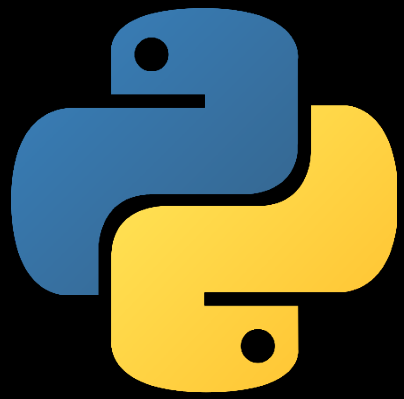
Current standing

Empire

Rebels

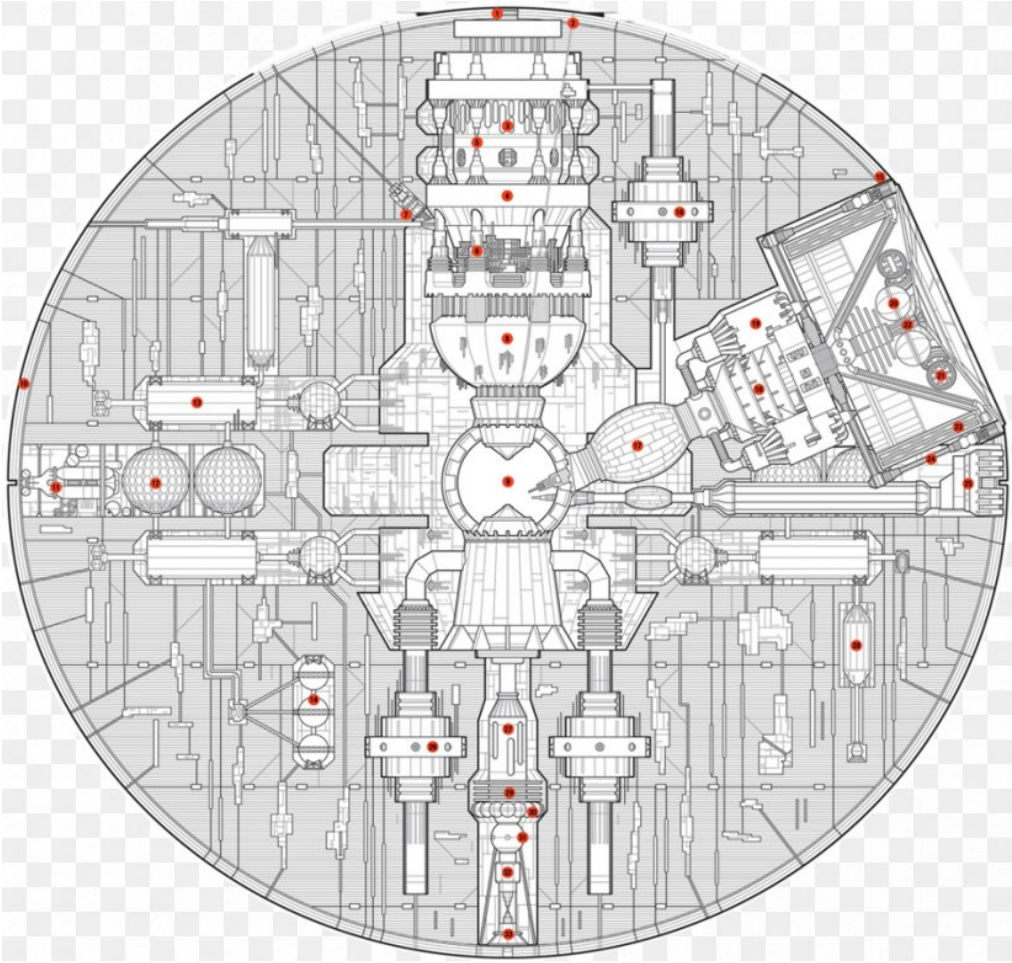
Build our web site, you will

THE FORCE
IS STRONG
WITH THIS ONE

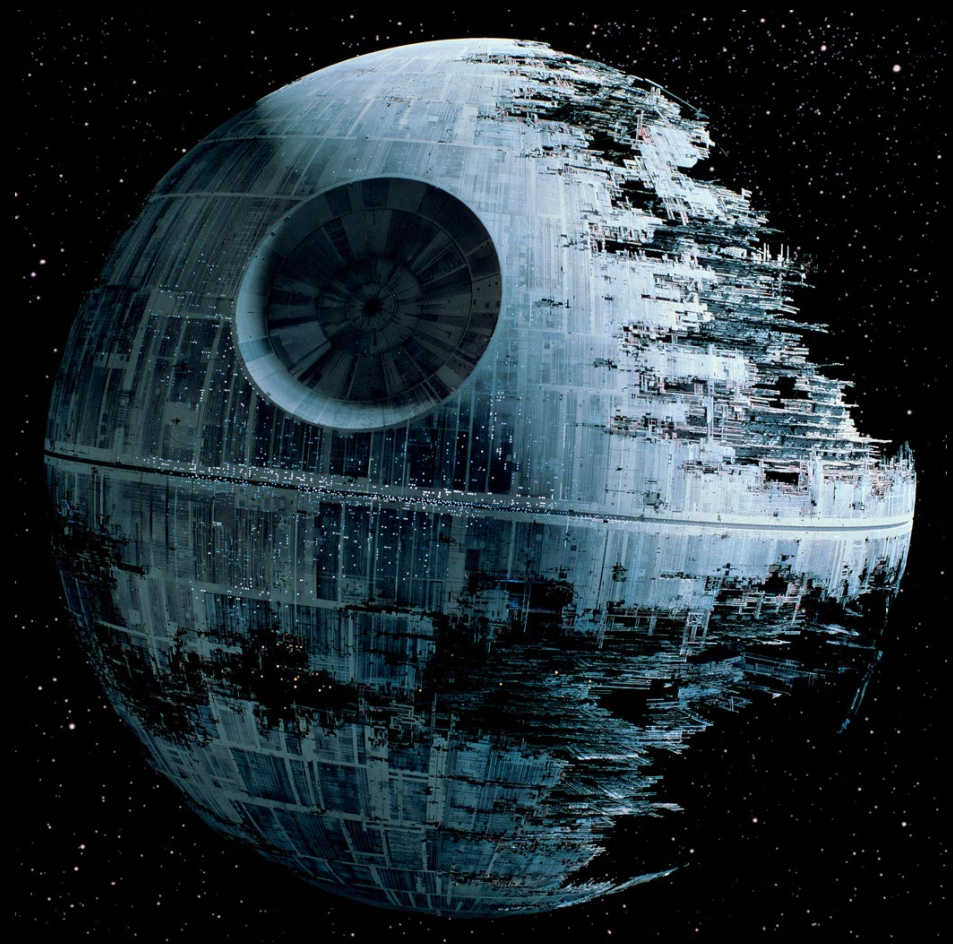


Dagens agenda

- Demo på vad vi ska bygga
- Bygga webbsidor med Python
  - Hur fungerar en webbserver?
  - Webbserver med Python
  - Server => Python
  - Klient => HTML/CSS
- Bottle som mikroramverk
  - Vad är ett mikroramverk?
  - Routes => Funktioner
- Bygga vår webbplats



Vad vi ska bygga



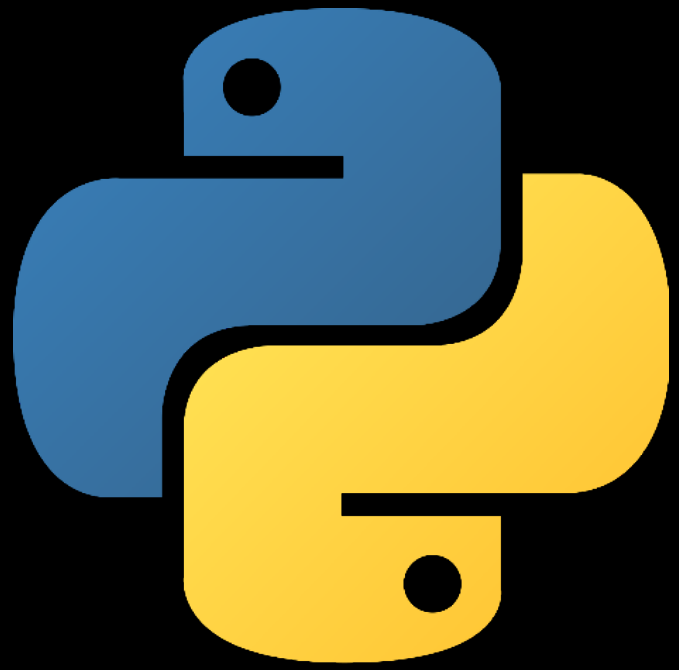
“You can’t stop the change, any more than you can stop the suns from setting.”

– Shmi Skywalker

Important, this is!

Webbserver?

En **webbserver** är antingen ett datorprogram som tillhandahåller webbsidor för en viss webbplats eller en serverdator på vilken sådan programvara körs.



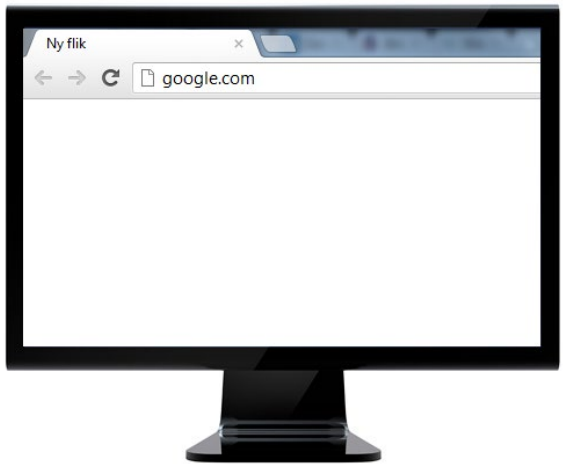
...datorprogram som tillhandahåller webbsidor ...

Internet

Webbserver

- Ett webbserverprogram har som uppgift att tillhandahålla webbsidor och andra filer via protokollet HTTP eller HTTPS, **vanligen över Internet**.
- Det förekommer också att **en enskild användare** kör en webbserver för eget lokalt bruk **på sin egen dator**. Vanligen kommunicerar användaren med webbservern med hjälp av en webbläsare.
- Användaren väljer webbsidor och webbläsaren *”beställer”* webbsidorna från webbservern och visar dem på användarens datorskärm. Webbläsaren är klient till webbservern.

Klient



Klient

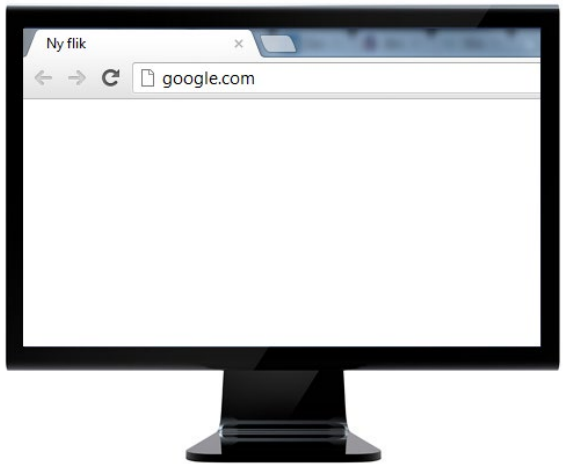
Förfrågan

Internet

Svar

Server

Klient



http://google.com

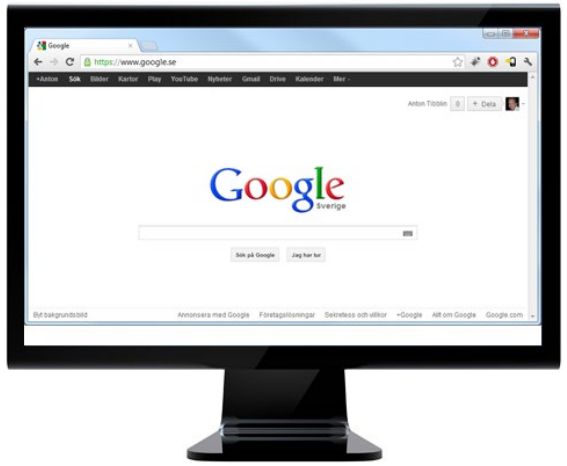
Förfrågan

Internet

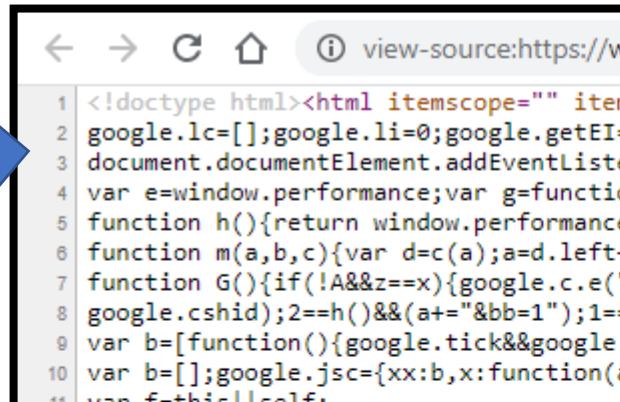
Server

Svar

Klient



Index.html



URL

Uniform Resource Locator

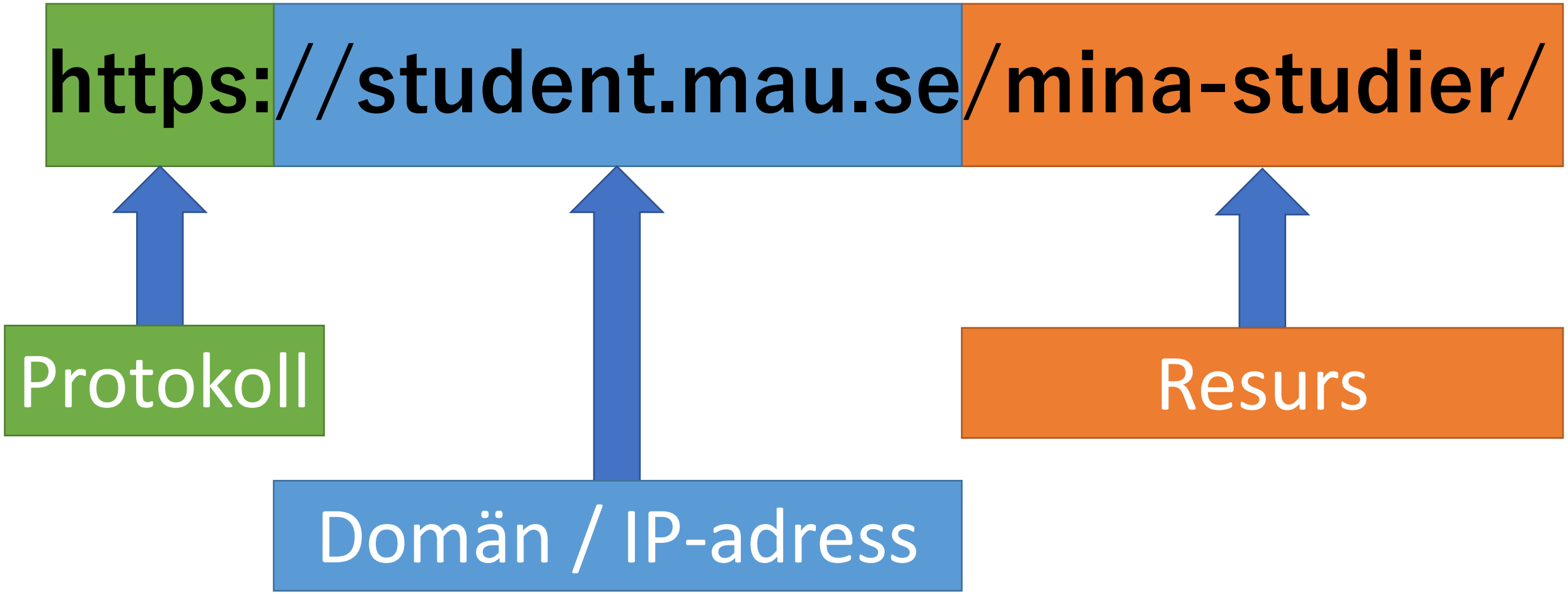
Exempel på URL

https://mau.se

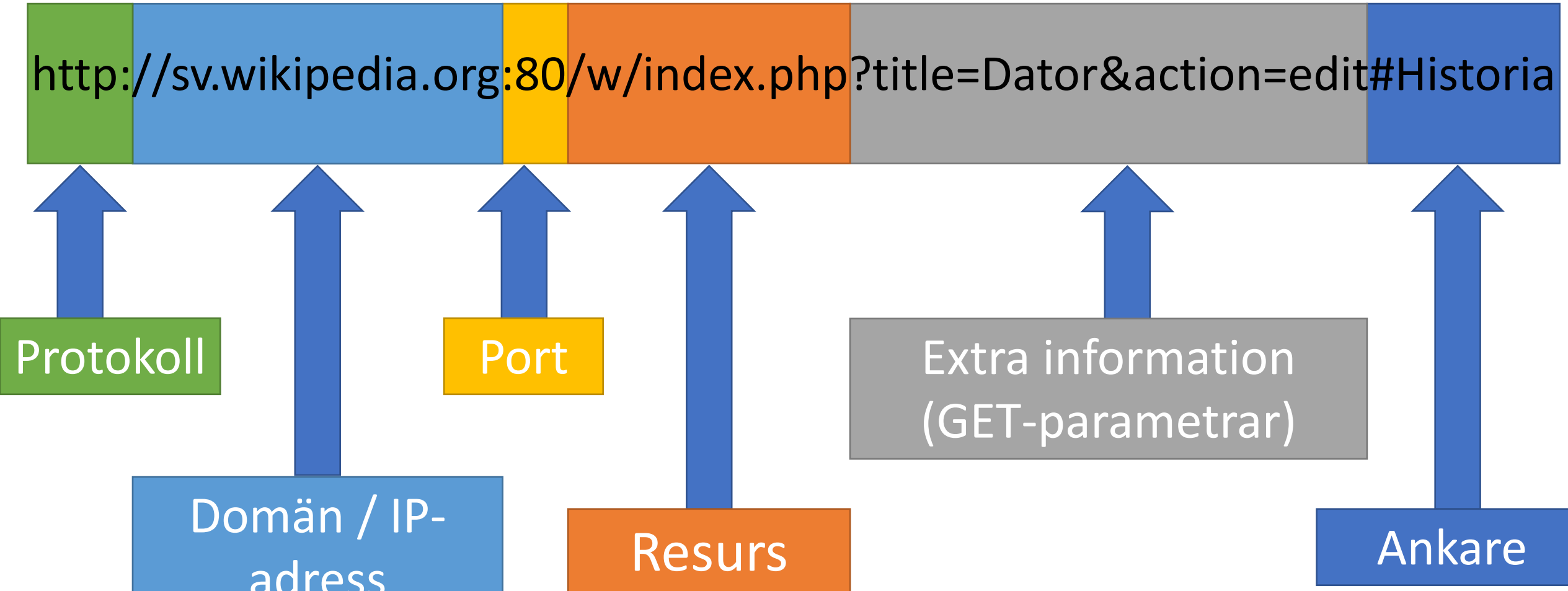
https://student.mau.se/mina-studier/

https://mau.instructure.com/courses/6601/discussion_topics/128343

Grundläggande URL



Komplett URL



Andra URI

Uniform Resource Identifier

ftp://ftp.is.co.za/rfc/rfc1808.txt

http://www.ietf.org/rfc/rfc2396.txt

mailto:John.Doe@example.com

telnet://192.0.2.16:80/

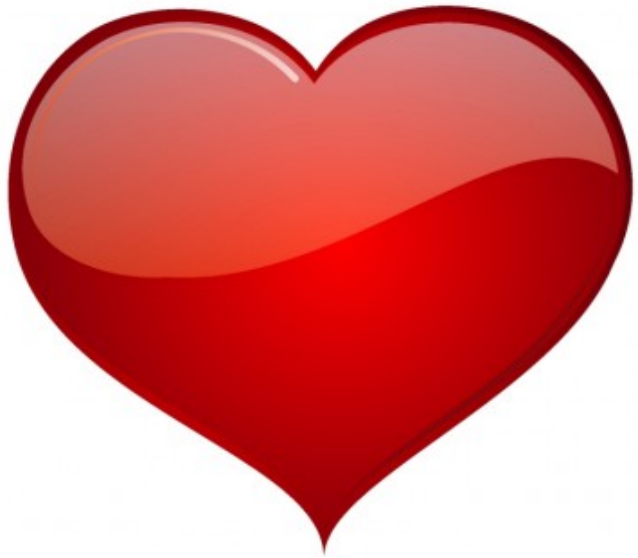
news:comp.infosystems.www.servers.unix

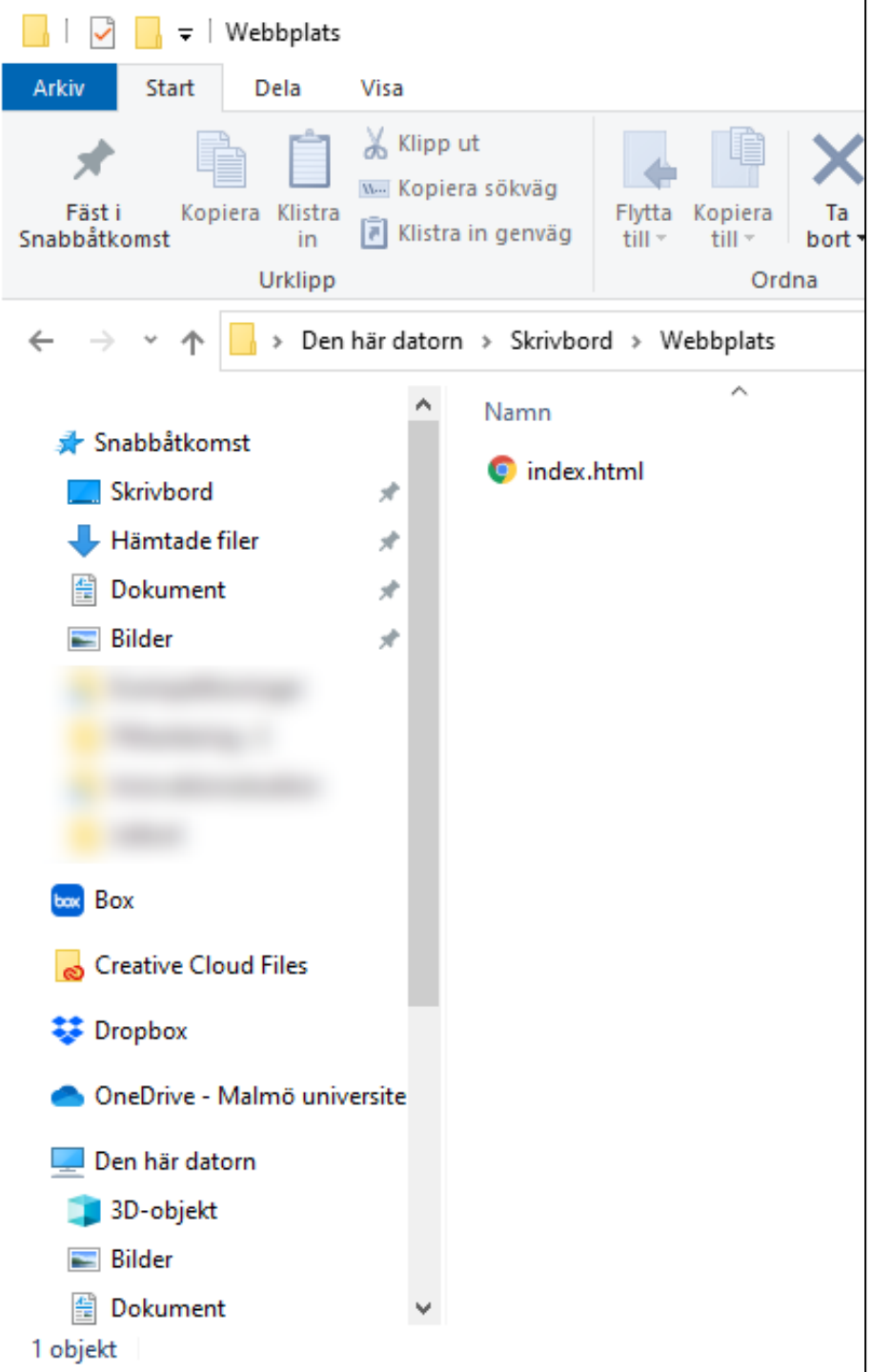
There's no place like 127.0.0.1

No place like 127.0.0.1, there is.

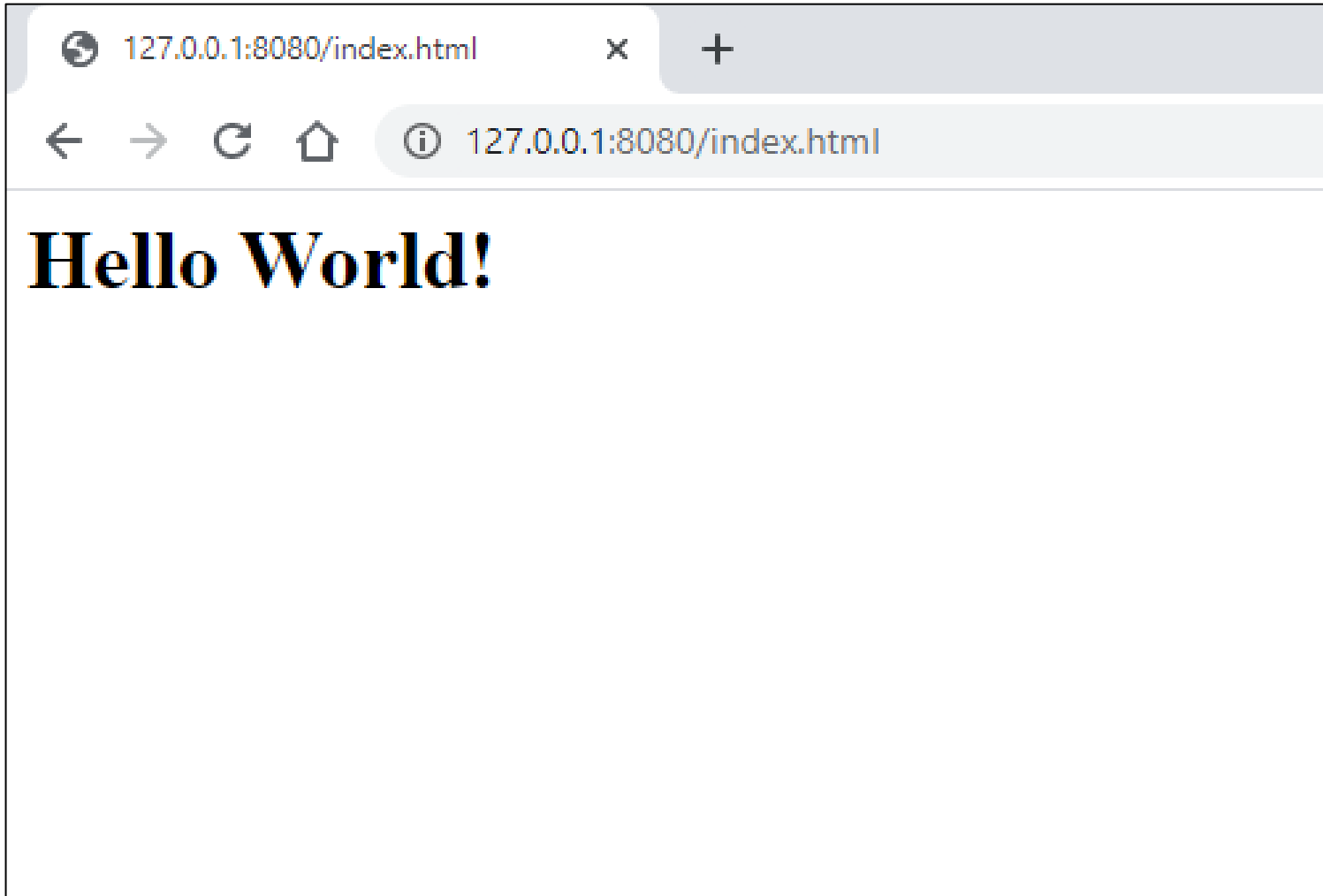
python -3 -m http.server 8000

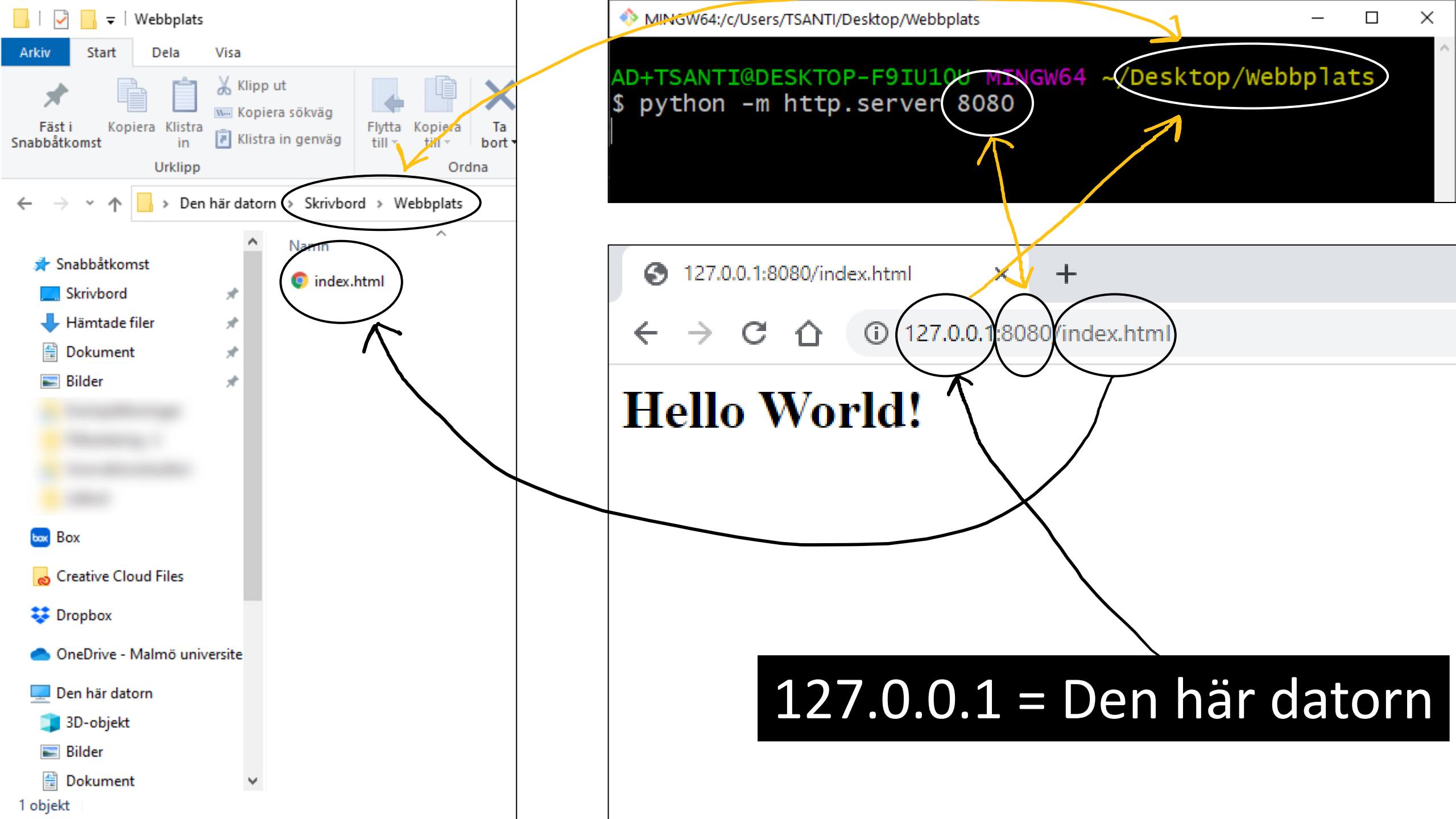
Starta en webbserver från en specifik mapp




Webbplats
Arkiv
Start
Dela
Visa
Fäst i Snabbåtkomst
Kopiera
Klistra in
Klipp ut
Kopiera sökväg
Klistra in genväg
Urklipp
Flytta till
Kopiera till
Ta bort
Ordna
Den här datorn
Skrivbord
Webbplats
Namn
index.html
Snabbåtkomst
Skrivbord
Hämtade filer
Dokument
Bilder
Box
Creative Cloud Files
Dropbox
OneDrive - Malmö universite
Den här datorn
3D-objekt
Bilder
Dokument
1 objekt

MINGW64:/c/Users/TSANTI/Desktop/Webbplats
AD+TSANTI@DESKTOP-F9IU10U MINGW64 ~/Desktop/Webbplats
$ python -m http.server 8080


127.0.0.1:8080/index.html
127.0.0.1:8080/index.html
Hello World!


Webbplats
Arkiv
Start
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Fäst i Snabbåtkomst
Kopiera
Klistra in
Klipp ut
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Klistra in genväg
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Flytta till
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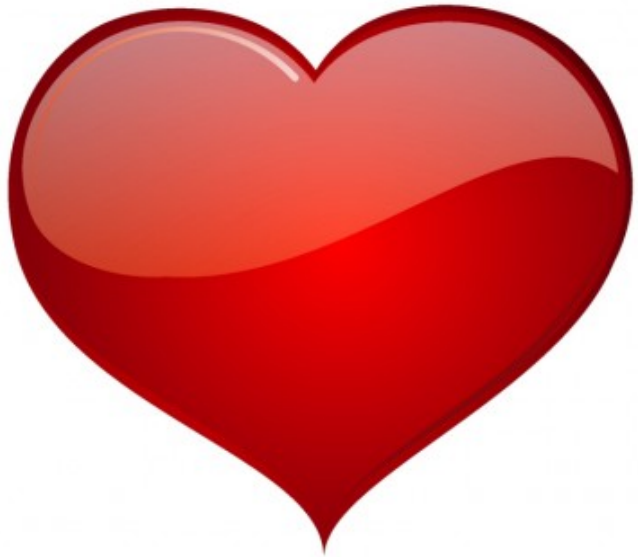
Ta bort
Ordna
Den här datorn
Skrivbord
Webbplats
Namn
index.html
Snabbåtkomst
Skrivbord
Hämtade filer
Dokument
Bilder
Box
Creative Cloud Files
Dropbox
OneDrive - Malmö universite
Den här datorn
3D-objekt
Bilder
Dokument
1 objekt
MINGW64:/c/Users/TSANTI/Desktop/Webbplats
AD+TSANTI@DESKTOP-F9IU10U MINGW64 ~/Desktop/Webbplats
$ python -m http.server 8080
127.0.0.1:8080/index.html
127.0.0.1:8080/index.html
Hello World!
127.0.0.1 = Den här datorn



Demo – Lokal webbserver

Men – hur mycket Python var detta egentligen?

Feel the power of the server side!

STAR WARS - VOTE

Please choose side

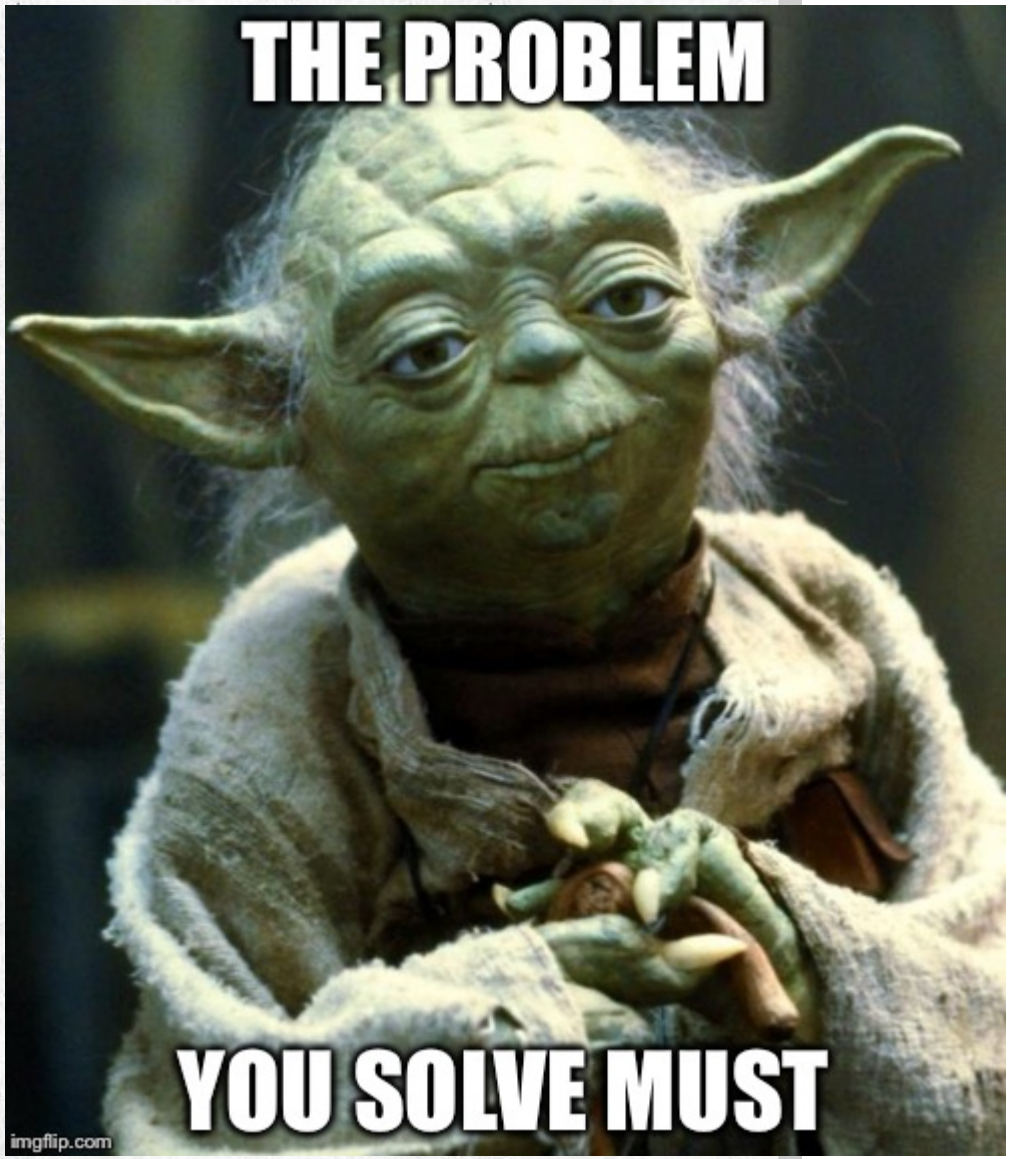
Empire

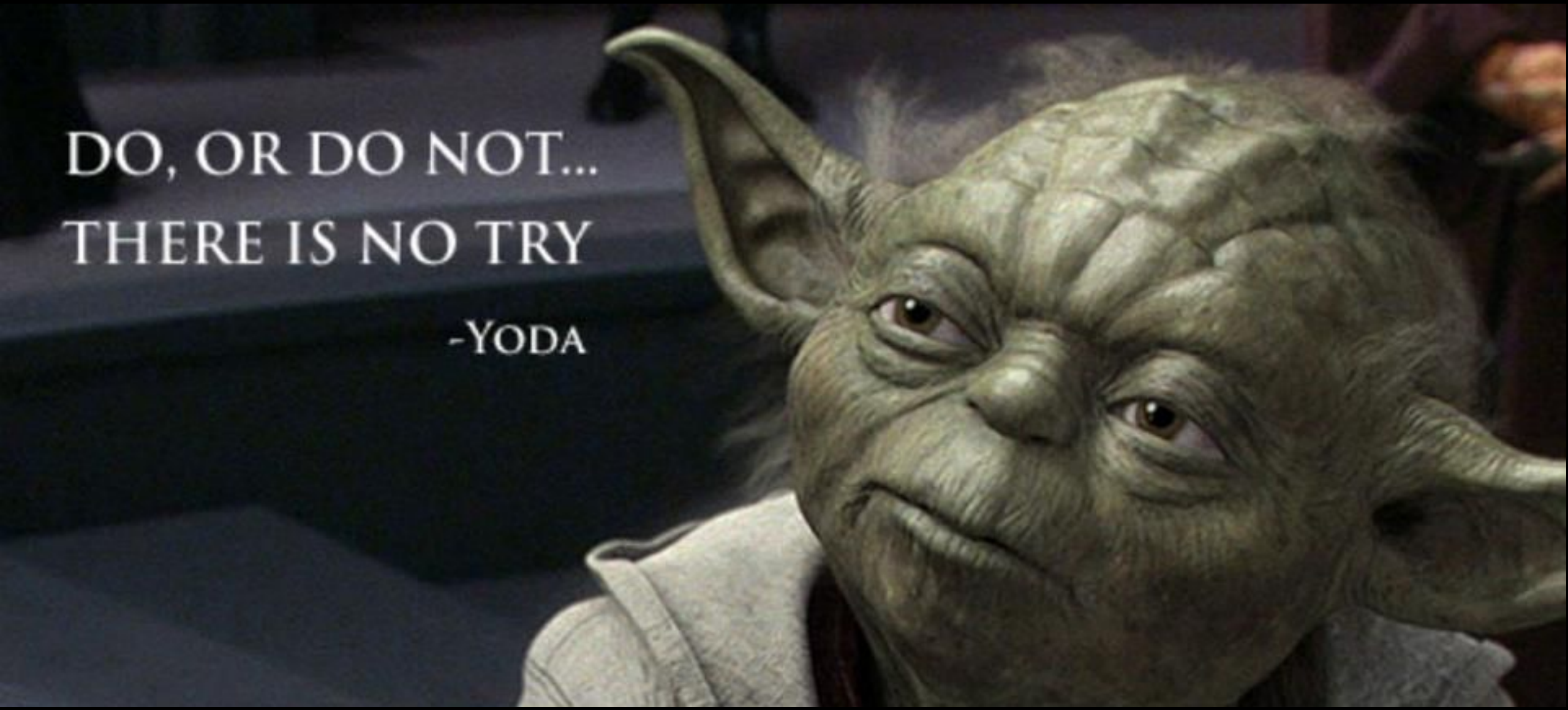
Rebels

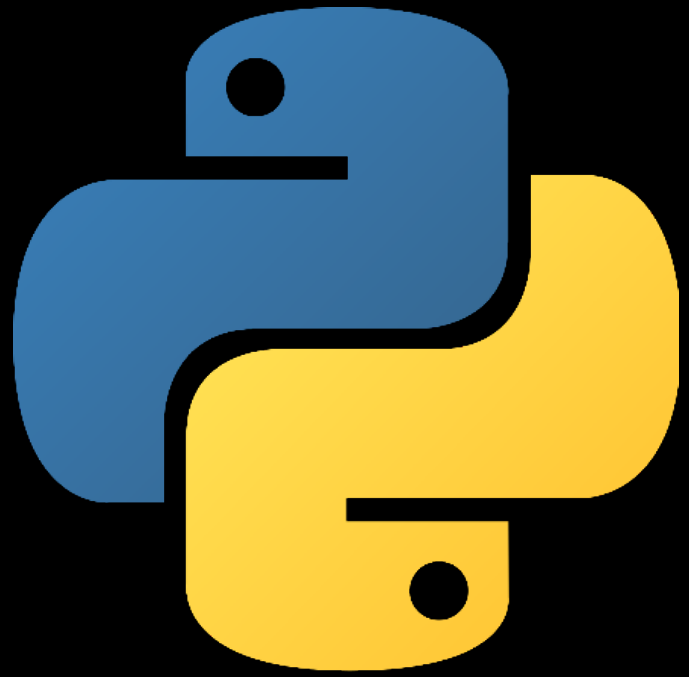
Vote

Current standing

Empire	Rebels
14	12




DO, OR DO NOT...
THERE IS NO TRY
-YODA



THE FORCE
IS STRONG
WITH THIS ONE

Vad kan vi hittills?

DA106A - Webbutveckling

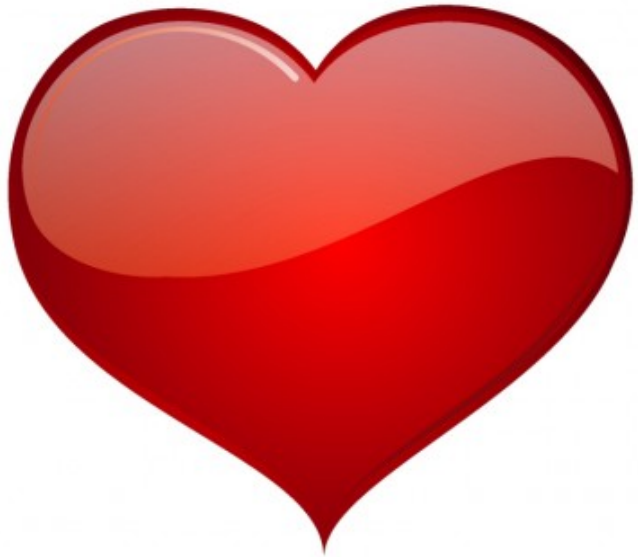
- Skapa webbsidor genom:
  - HTML
  - CSS
  - (JavaScript)

DA354A - Programmering

- Programmera i Python
  - Variabler
  - Datatyper
  - Funktioner
  - Filhantering
  - Felhantering
  - JSON
  - M.m.

Vad behöver vi lära oss? => Bygga en webbtjänst!

“Oh, my dear friend. How I’ve missed you.”
- C-3PO



http://www.

Behöver vi bygga allt själva?

A simpler way, there is.

To be efficent, a tool you need.

https://www.mindinventory.com/blog/best-python-web-frameworks-2019/

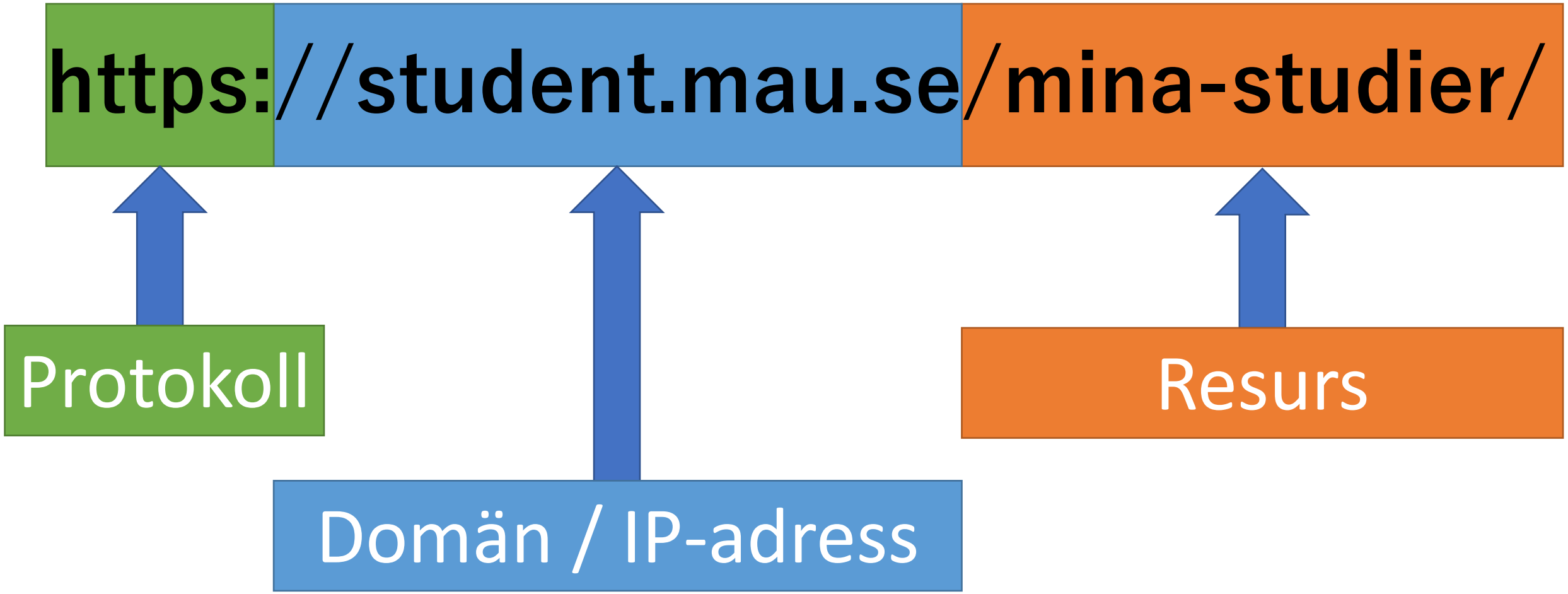
Bottle

“I’m just a simple man trying to make my way in the universe.”

- Jango Fett

Koncept 1
- Routes

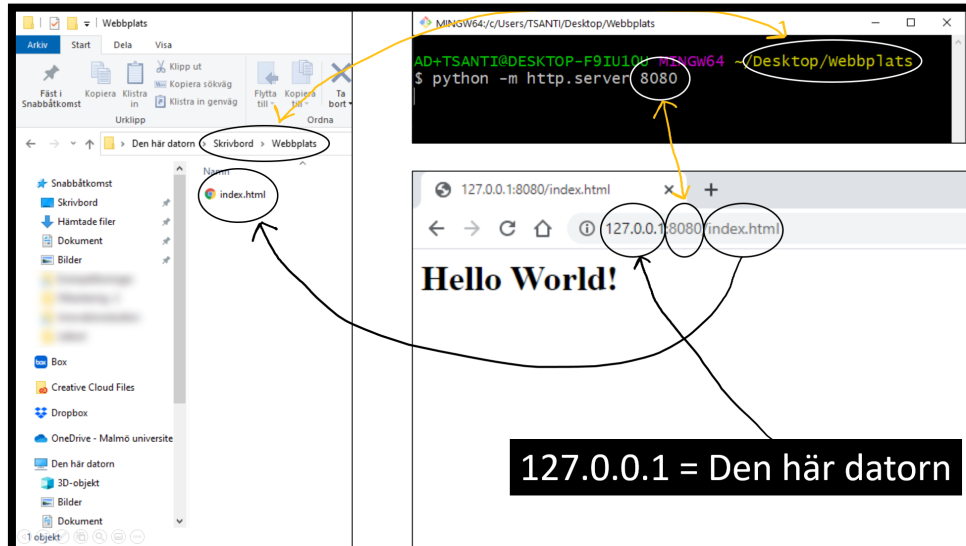
Grundläggande URL



Bottle - Routes

- Vi vill kunna mappa URL till olika funktioner i Python
  - /home => def home()
  - /contact => def contact()
  - /about => def about()

- 127.0.0.1/home => def home()
- 127.0.0.1/contact => def contact()
- 127.0.0.1/about => about()



Kom igång med routes

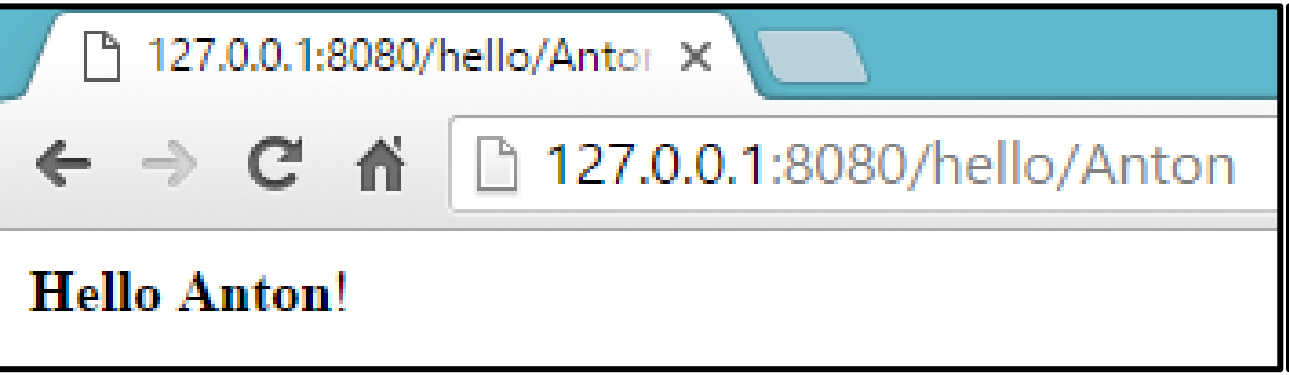
“
I'M ONE WITH THE FORCE, AND THE FORCE IS WITH ME.
CHIRRUT ÎMWE
MAGICALQUOTE

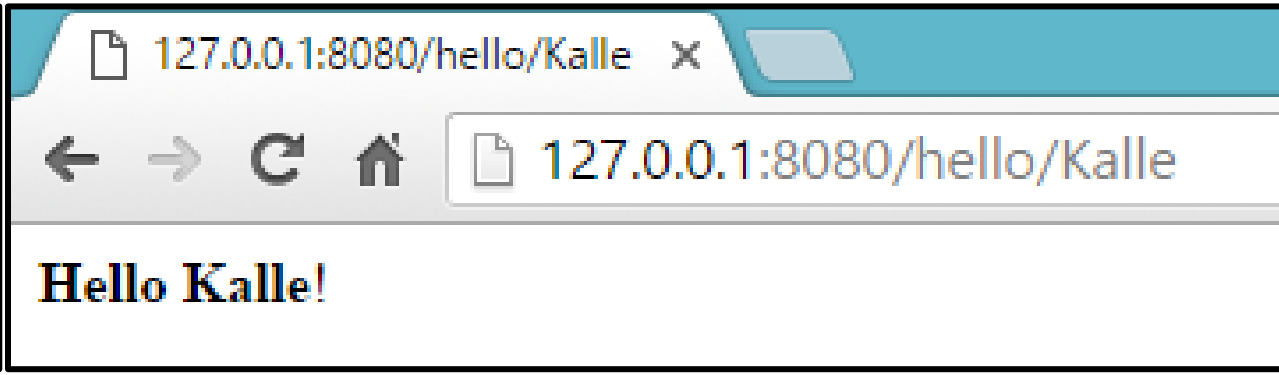
Hello <name>

```
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```





```python
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@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```
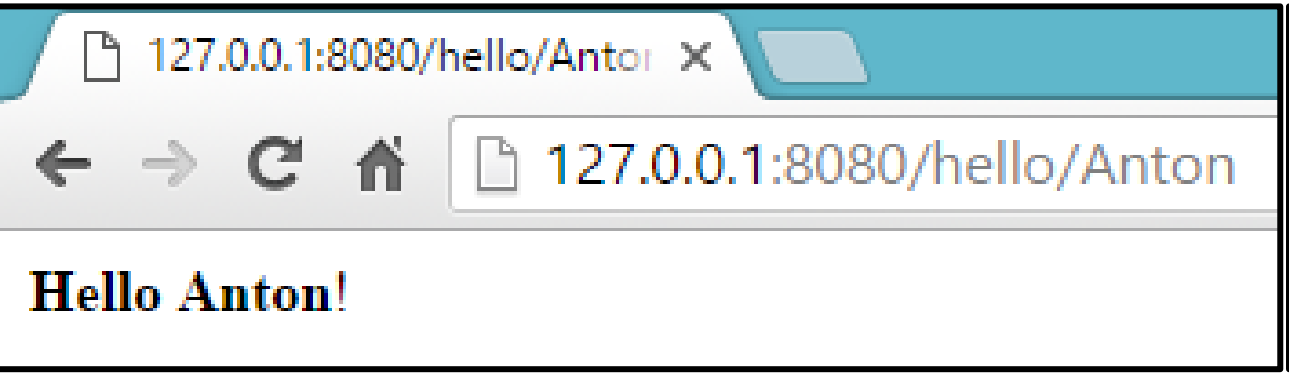
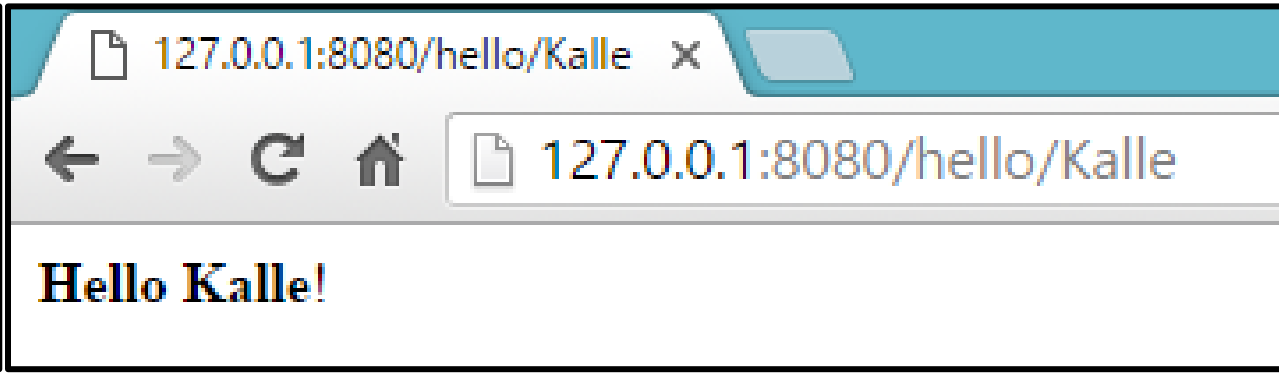



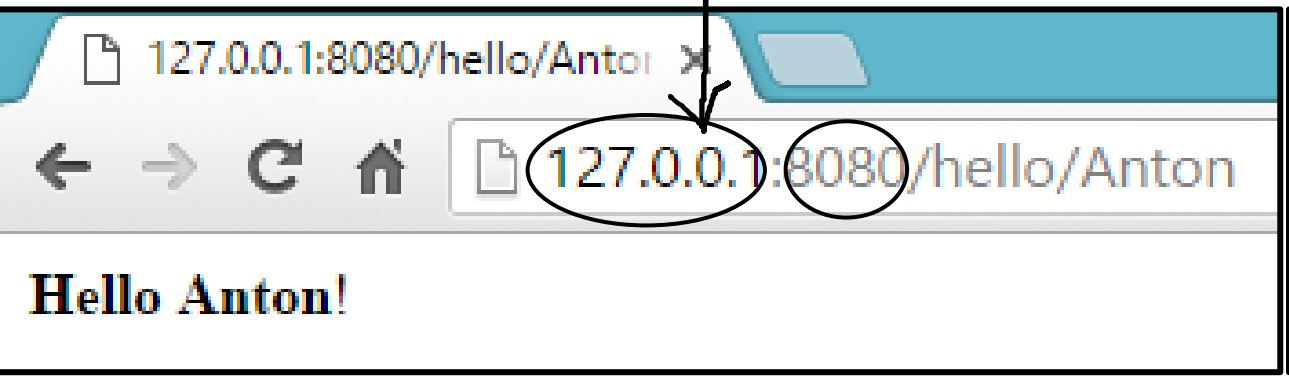


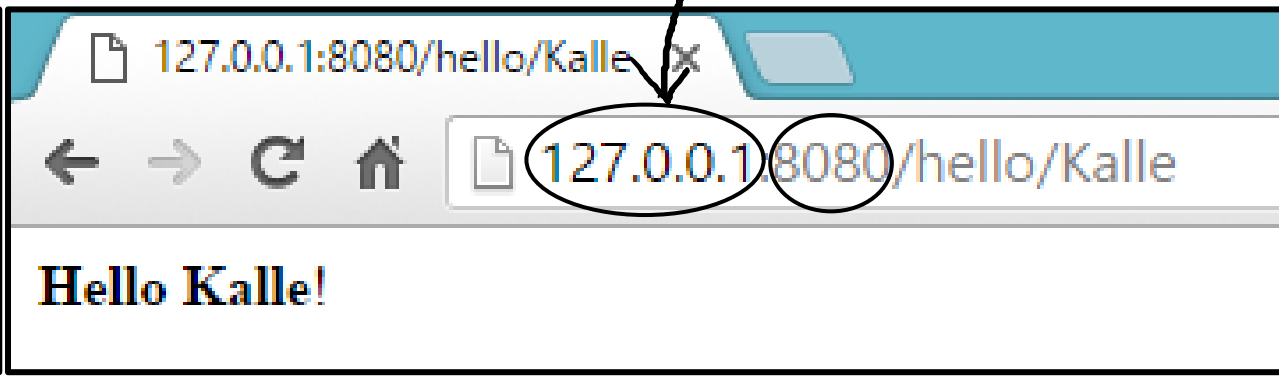
1. Första grunderna i koden

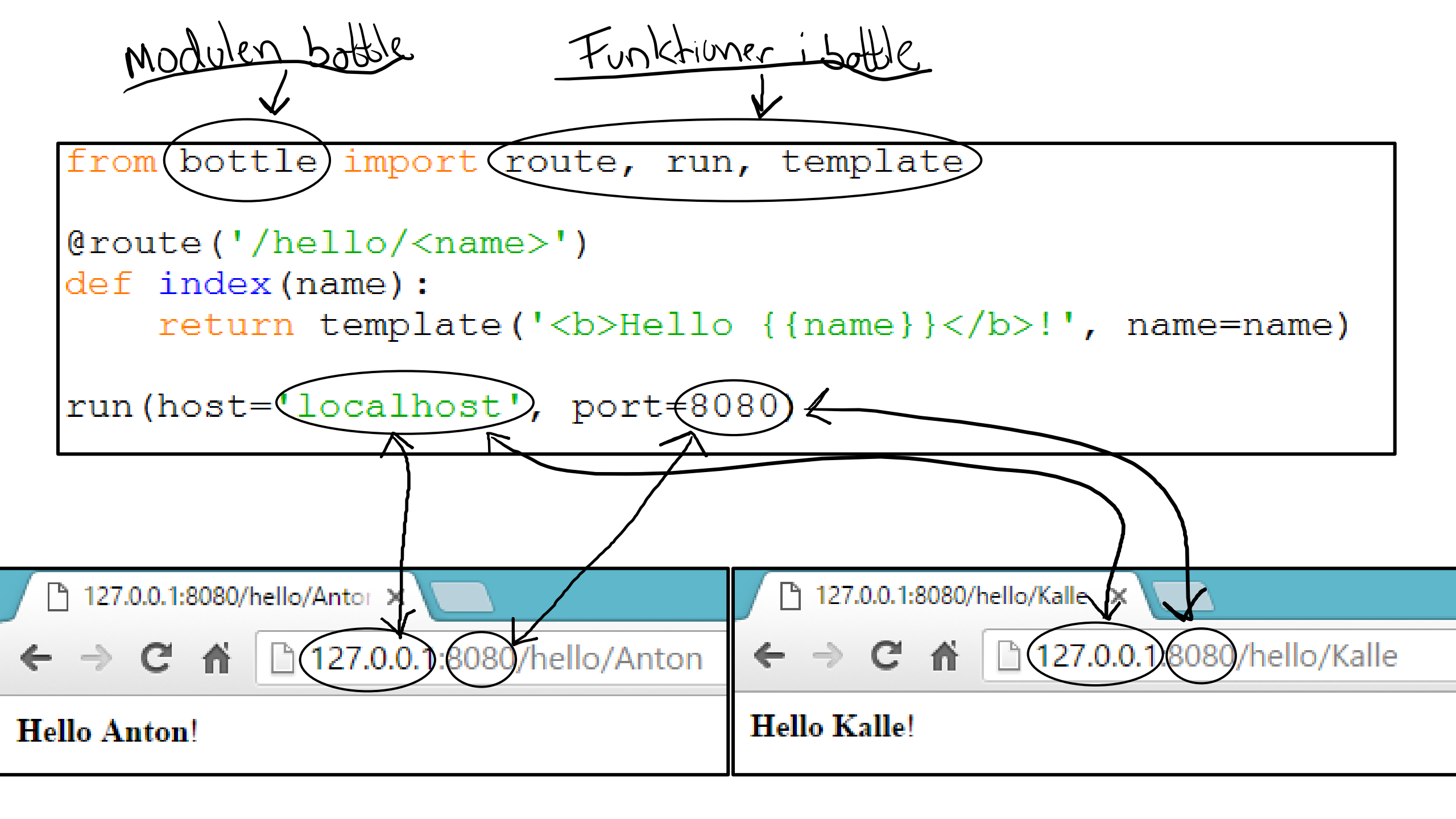
Modulen bottle

Funktioner i bottle

```
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```





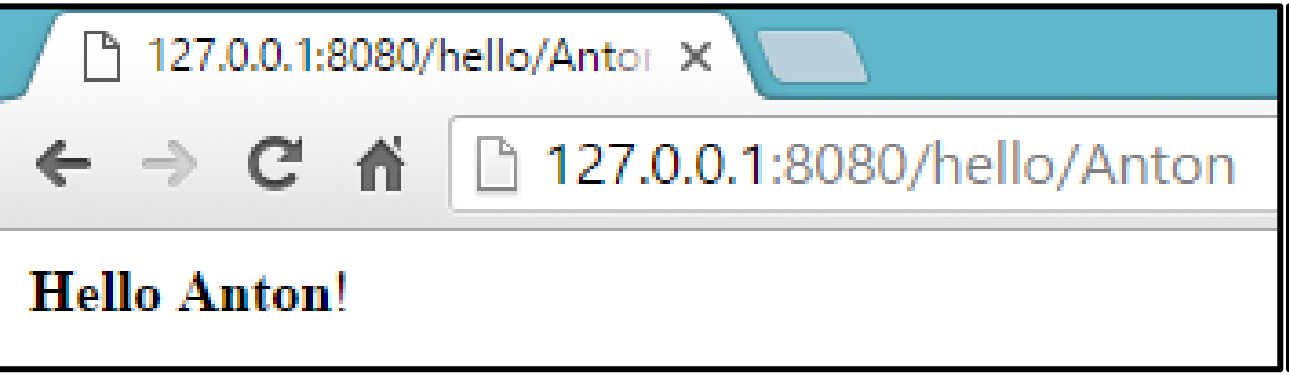

Modulen bottle
Funktioner i bottle
from bottle import route, run, template
@route('/hello/<name>')
def index(name):
return template('Hello {{name}}!', name=name)
run(host='localhost', port=8080)
127.0.0.1:8080/hello/Anton
Hello Anton!
127.0.0.1:8080/hello/Kalle
Hello Kalle!

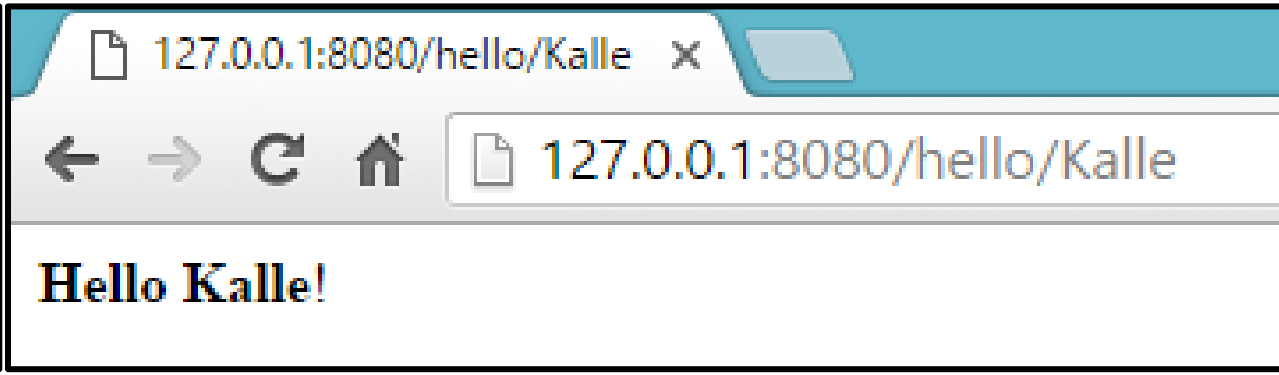
2. Förstå routes och argument

```python
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```





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@route('/hello/<name>')
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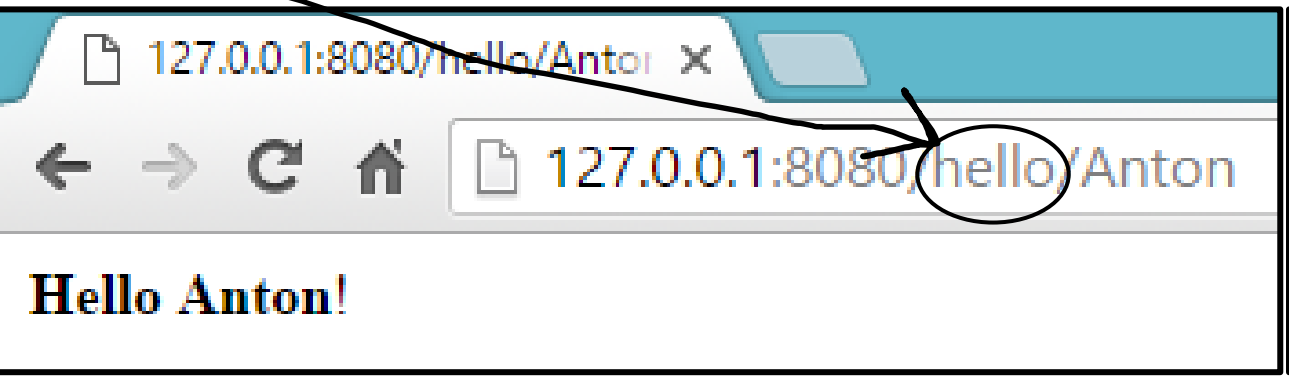
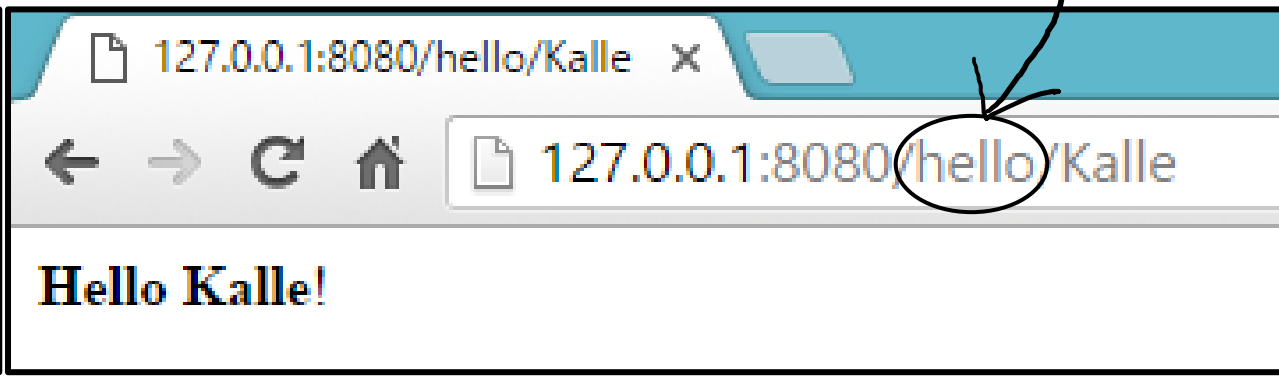
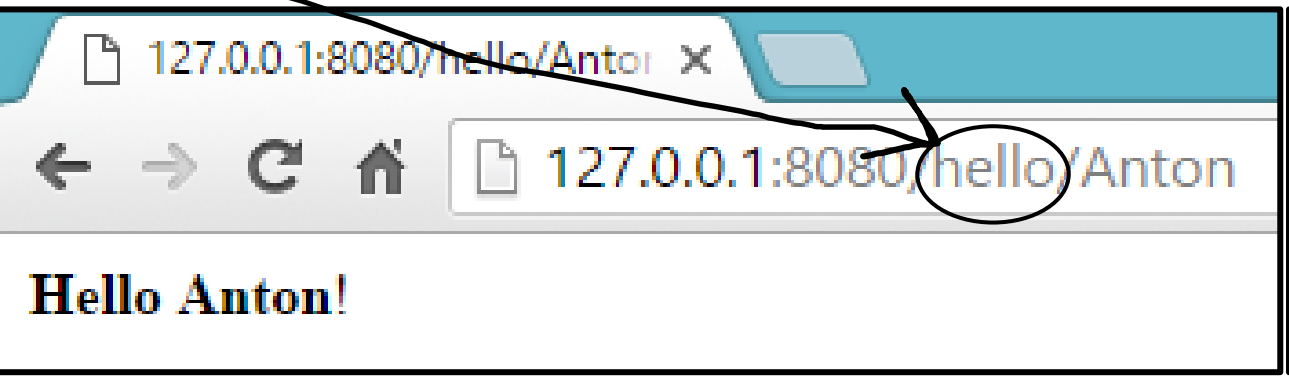
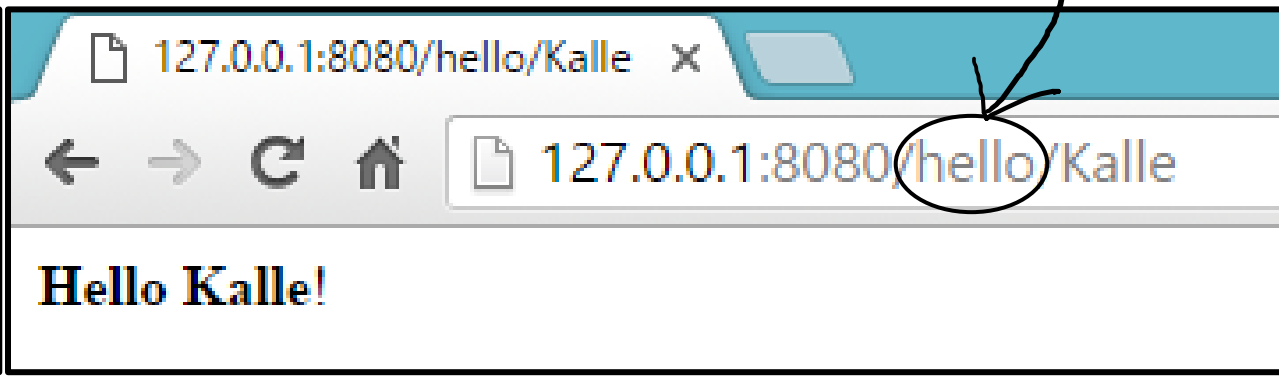
```

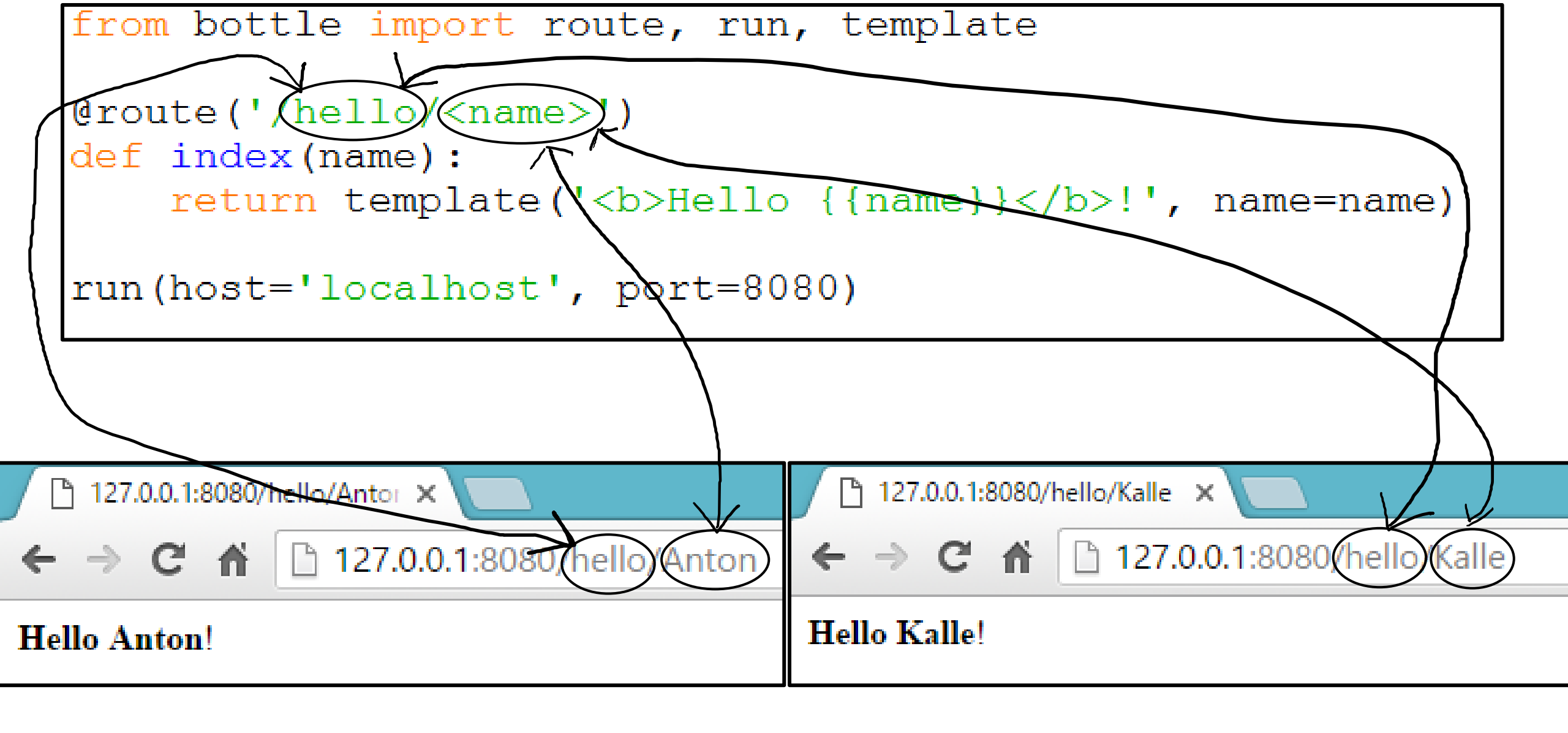




```python
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```

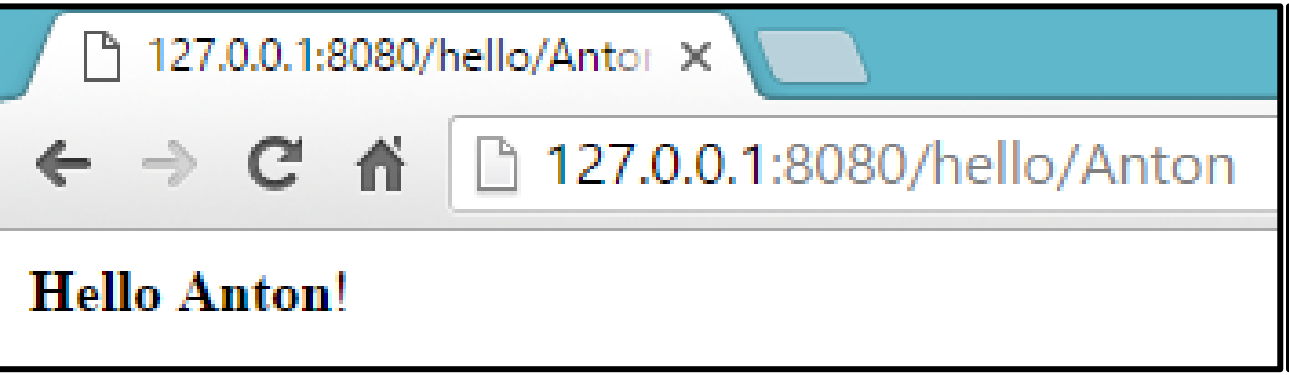


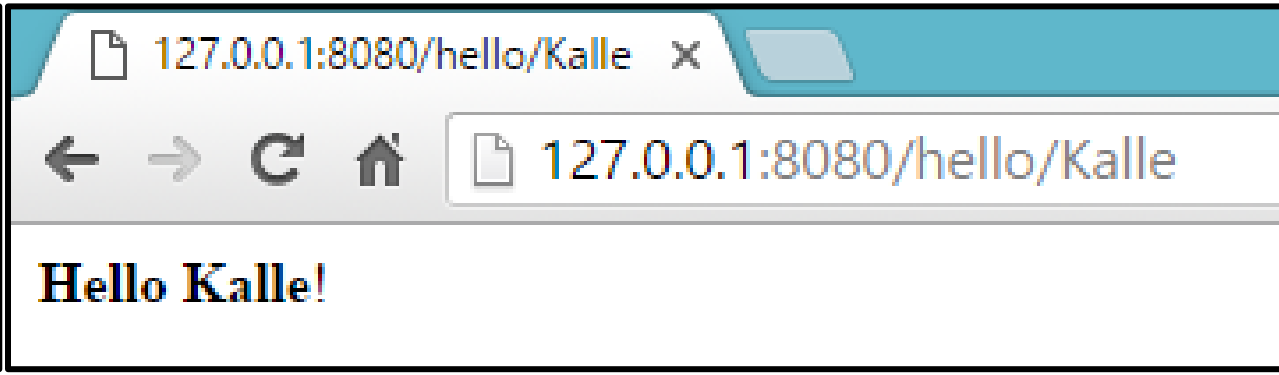



from bottle import route, run, template
@route('/hello/<name>')
def index(name):
return template('Hello {{name}}!', name=name)
run(host='localhost', port=8080)
127.0.0.1:8080/hello/Anton
Hello Anton!
127.0.0.1:8080/hello/Kalle
Hello Kalle!

3. Förstå parametrar (& värde)

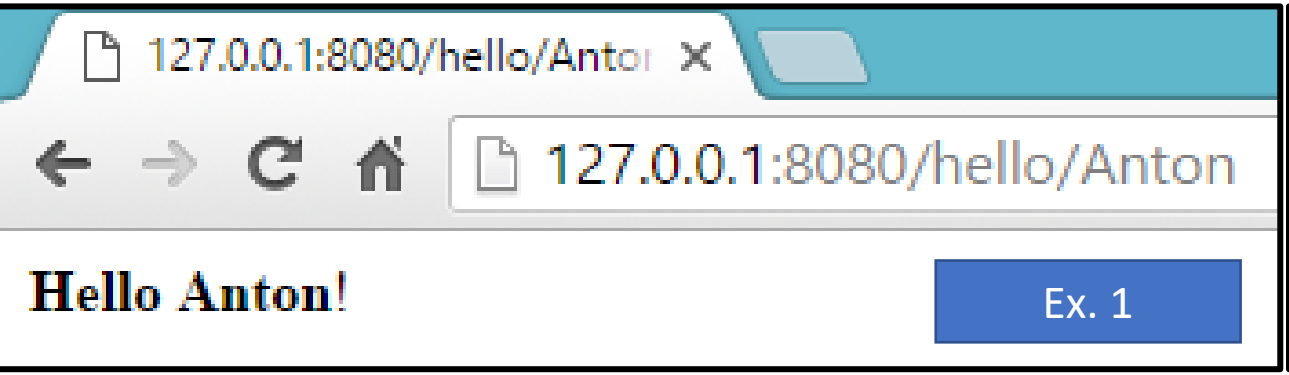
```python
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```

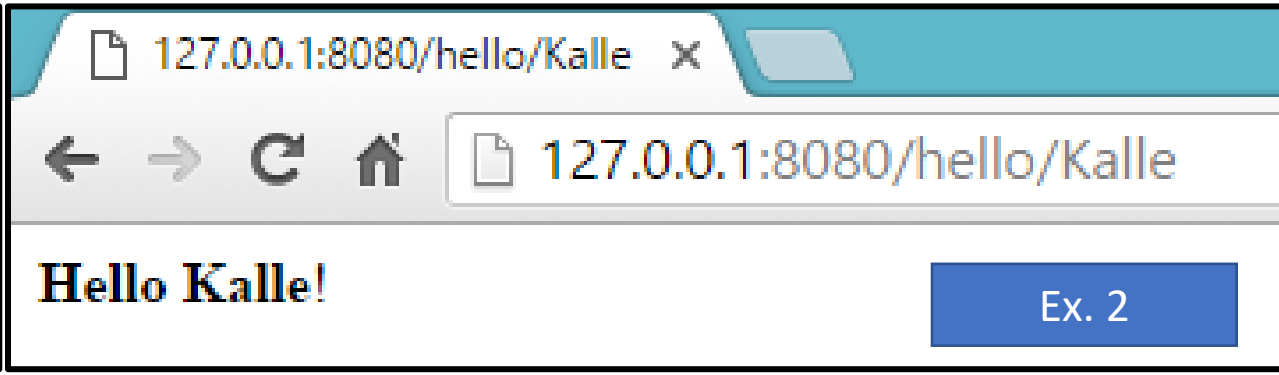




```python
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```

Ex 1 - Anton
Ex 2 - Kalle

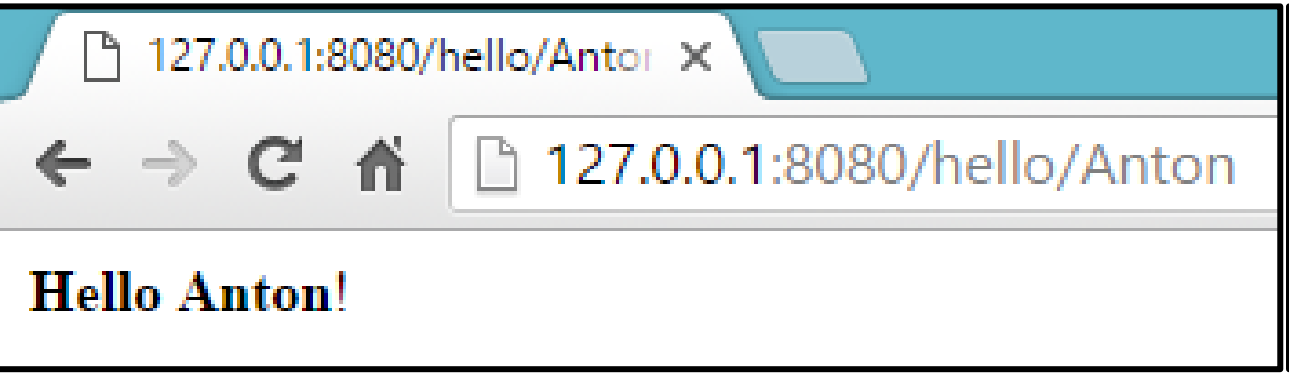


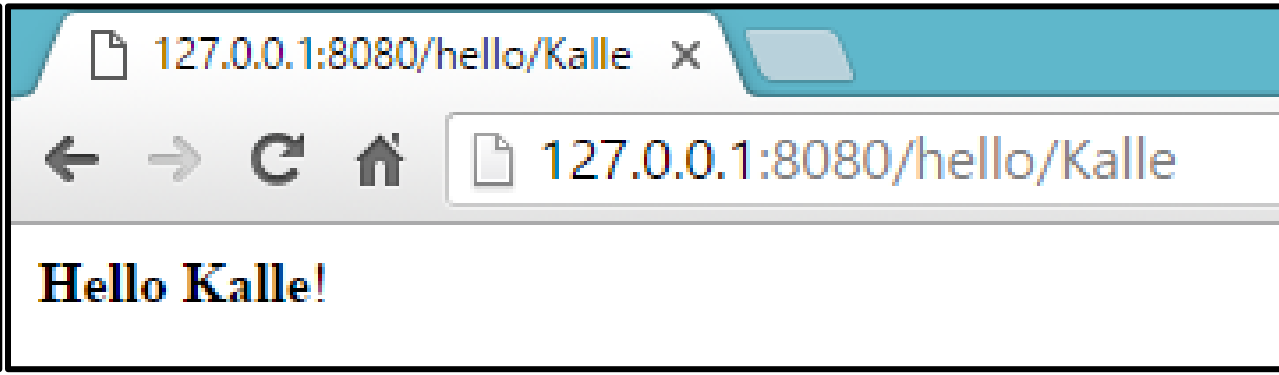


```
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```

Ex 1 - Anton
Ex 2 - Kalle

Namn på parameter till template

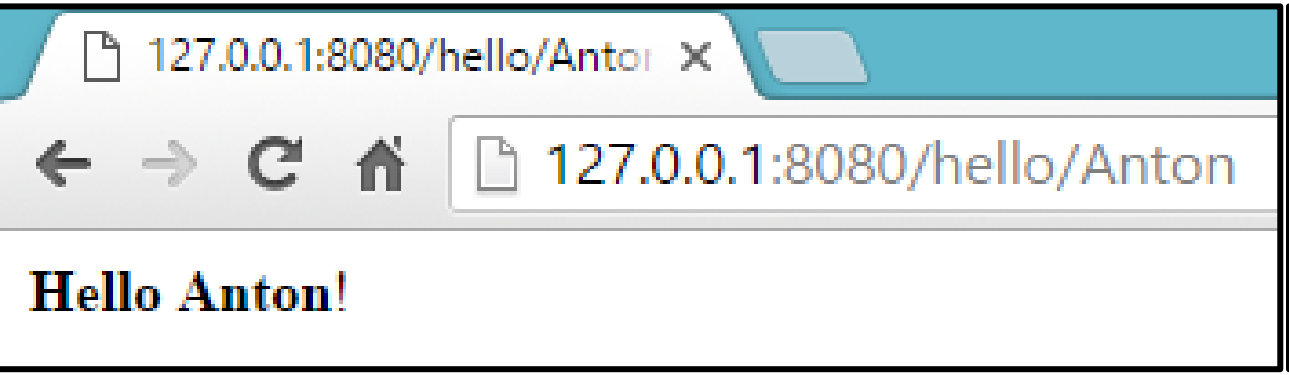


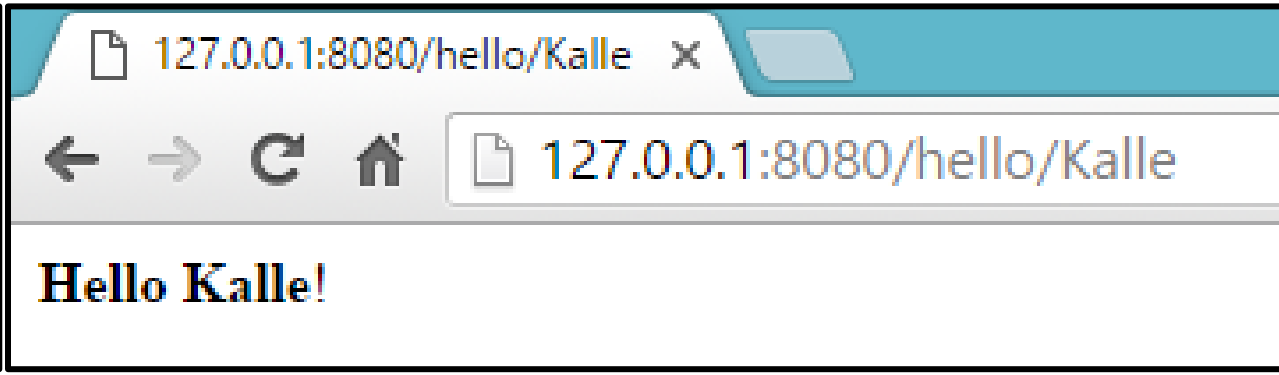


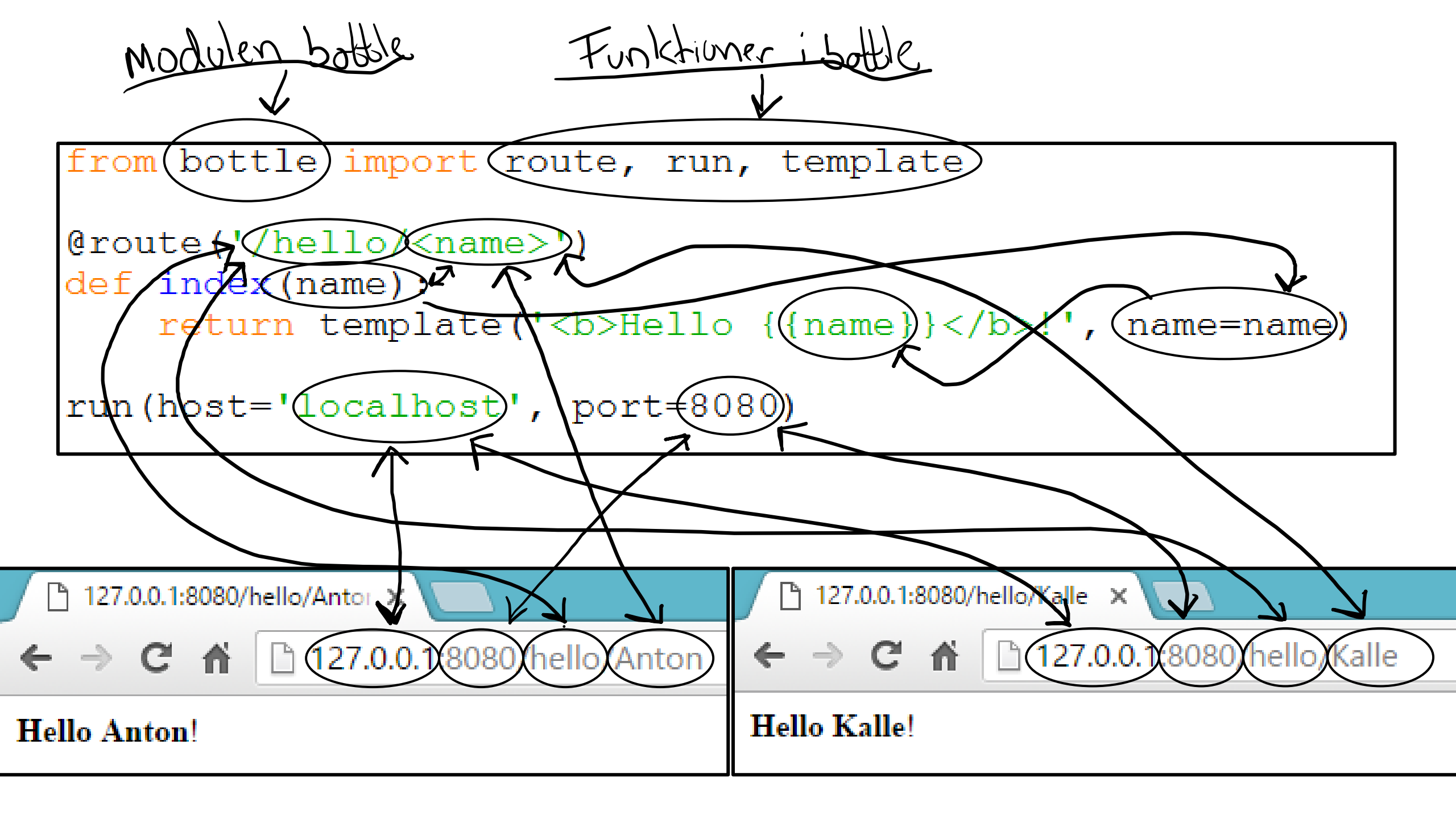
- En total översikt

Riskerar att bli lite förvirrande ´ …

```python
from bottle import route, run, template

@route('/hello/<name>')
def index(name):
    return template('<b>Hello {{name}}</b>!', name=name)

run(host='localhost', port=8080)
```



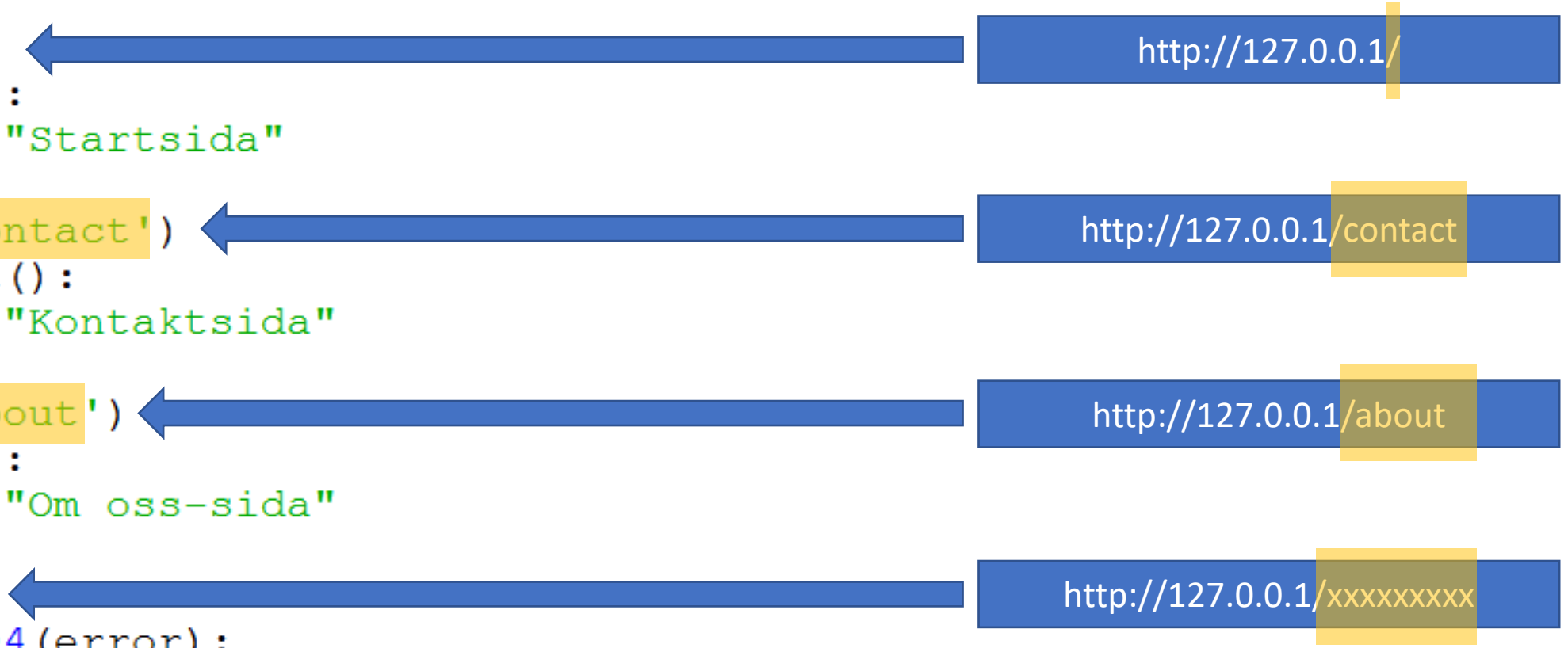



Modulen bottle
Funktioner i bottle
from bottle import route, run, template
@route('/hello/<name>')
def index(name):
return template('Hello {{name}}!', name=name)
run(host='localhost', port=8080)
127.0.0.1:8080/hello/Anton
Hello Anton!
127.0.0.1:8080/hello/Kalle
Hello Kalle!

THIS IS THE WAY

Att använda sig utav flera routes...

```python
from bottle import route, run, error

@route('/')
def index():
    return "Startsida"

@route('/contact')
def contact():
    return "Kontaktsida"

@route('/about')
def about():
    return "Om oss-sida"

@error(404)
def error404(error):
    return "Denna sida finns inte..."

run(host='localhost', port=8080)
```



DEMO - Routes

Koncept 2
- Templates

Templates – Mallar i HTML

Join the dark side, we have the prettiest web sites!

Bottle - Templates

- Templates är mallar för hur vi ska presentera vår information
  - HTML
  - CSS
  - JavaScript

- Vi kan skicka data (bearbetad utav python) till våra malla för att skapa dynamiska webbsidor

Bottle – Templates och statiska filer

- Alla templates ska:
  - Ligga i mappen ”views”
  - Ha filändelsen *.tpl eller *.html

- Statiska filer (bilder, css, js, etc.) ska:
  - Ligga i mappen ”static”

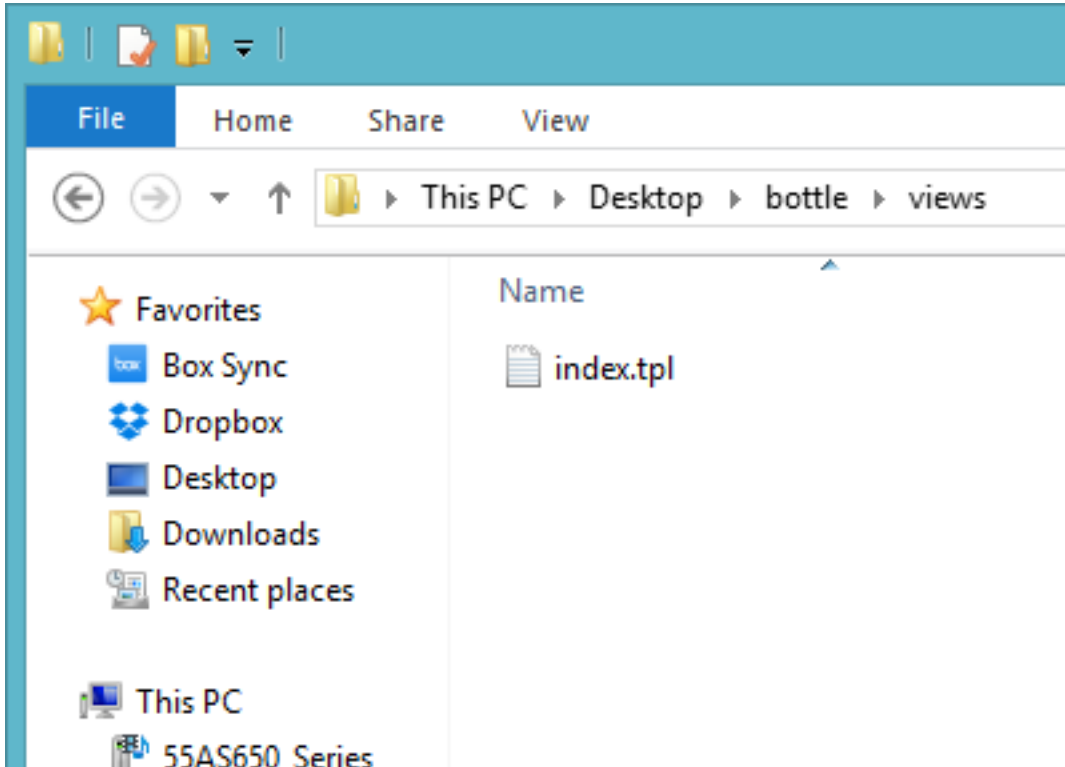
Hur ser detta ut i bottle?

bottle_app.py - C:\Users\TSANTI\Desktop\Bottle-templates\

File Edit Format Run Options Windows Help

```
from bottle import route, get, post, run, template,

@route('/')
def index(user_name):
        return template("index")

@route('/static/<filename>')
def server_static(filename):
    return static_file(filename, root='static')


@error(404)
def error404(error):
    return "Denna sida finns inte..."

run(host='localhost', port=8080)
```

```
<!doctype html>
<html>
    <head>
        <title>Star Wars</title>
        <meta charset="utf-8">
        <link href="/static/style.css" rel="stylesh
    </head>
    <body>
        <h1>Star Wars!</h1>
        <footer>
            Made by the force
        </footer>
    </body>
</html>
```



Demo - Templates

{JSON}
JavaScript Object Notation

JSON

JavaScript Object Notation

`json.dumps()` => omvandla datatyper till sträng

`json.loads()` => omvändla stäng till datatyp

Lista

```
[
    {
        "title": "Star Wars",
        "year": 1977,
        "director": "George Lucas"
    },
    {
        "title": "Fight club",
        "year": 1999,
        "director": "David Fincher"
    }
]
```

json.dumps

Sträng

```
”[
    {
        ’title': 'Star Wars',
        'year': 1977,
        'director': 'George Lucas'
    },
    {
        'title': 'Fight club',
        'year': 1999,
        'director': 'David Fincher'
    }
]”
```

Lista

json.loads

Sträng

Nu till vårt projekt

“There’s always a bigger fish.”
- Qui-Gon Jinn

libbelit.pythonanywhere.com

STAR WARS - VOTE

Please choose side

Empire

Rebels

Vote

Current standing

Empire

Rebels

Routes => Vilka behöver vi?

Route	Syfte	Metod	Template
/	Till vår startsida	GET	index
/vote	Rösta på en sida	POST	index
/disqus	Visa gästbok	GET	disqus
/new-post	Skriv ett inlägg	POST	disqus
error404	En sida som ej finns	Alla	error

Untitled

File Edit Format Run Options Windows Help

```
from bottle import route, get, post, run, template, error, static_file, request

@route('/')
def index():
    # View start page
    return template("index")

@route('/vote', method='POST')
def vote():
    # Save vote and view start page
    return template("index")

@route('/disqus')
def disqus():
    # View disqus page
    return template("disqus")

@post('/disqus-post', method='POST')
def save_post():
    # Save new disqus post and view disqus page
    return template("disqus")

@route('/static/<filename>')
def server_static(filename):
    return static_file(filename, root='static')

@error(404)
def error404(error):
    return template("error")

run(host='localhost', port=8080, debug=False)
```

Routes => Vilka behöver vi?

Route	Syfte	Metod	Template
/	Till vår startsida	GET	index
/vote	Rösta på en sida	POST	index
/disqus	Visa gästbok	GET	disqus
/new-post	Skriv ett inlägg	POST	disqus
error404	En sida som ej finns	Alla	error

Testa så att allt fungerar

Let the battle begin

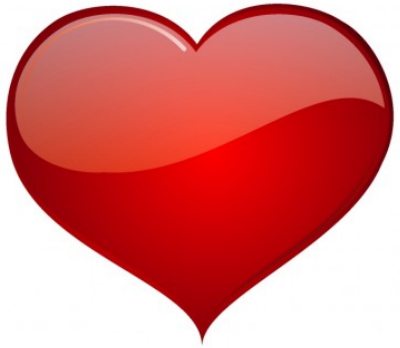
1. Fixa röstningen

2. Fixa gästboken

Röstningen

- Spara alla röster i filen ”votes.json”
- Spara rösterna som JSON-format, enligt mallen:

```
{
  "empire": 0,
  "rebels": 0
}
```



- Vi kan omvandla lexikon i Python till JSON genom:
  - json_votes = json.dumps(votes)
- Vi kan omvandla JSON till Python-datatyper genom:
  - votes = json.loads(json_votes)

{JSON}
JavaScript Object Notation

Flödesschema

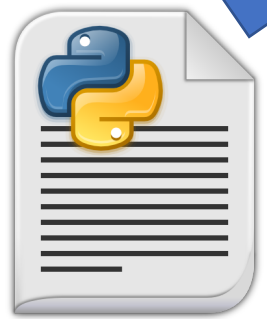
```
votes["empire"]
```

2

```
votes = json.loads(votes_file.read())
```

Läs in textfilens innehåll

Tolka JSON-data till ett lexikon



Skriv ut imperiets röster

3

Skriv ut Rebellers röster

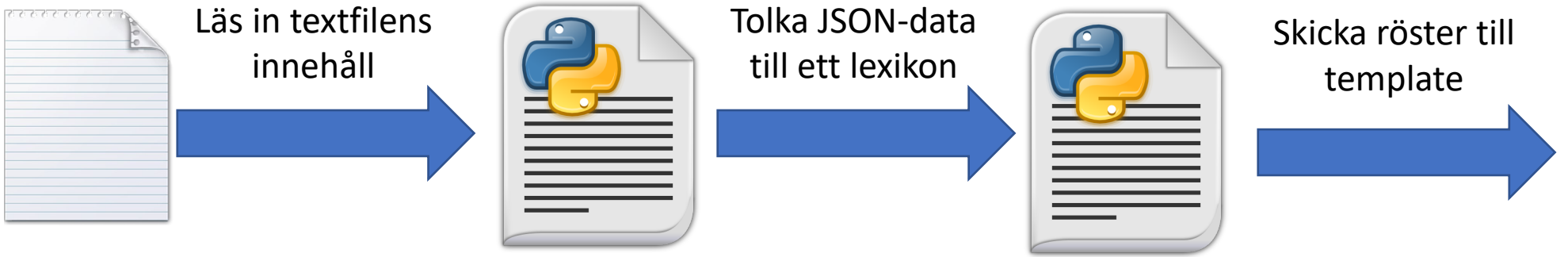
```
votes_file = open('static/standings.txt', 'r')
```

1

```
votes["rebels"]
```

```
{
    "empire": 0,
    "rebels": 0
}
```

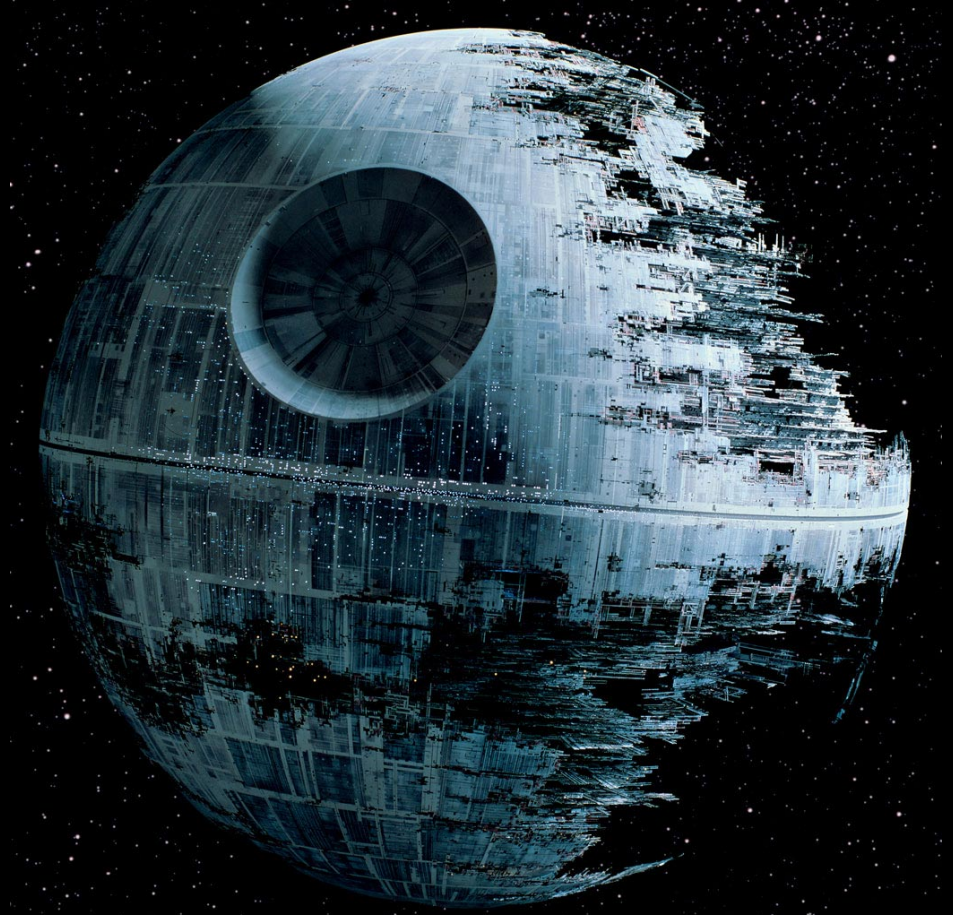
Flödesschema



Webbsida på internet

Skriv ut imperiets röster

Skriv ut Rebellers röster



Nu bygger vi! ;)

2. Fixa gästboken

“Power! *Unlimited* power!”
- Darth Sidious

Frågor?